

BFS Traversal

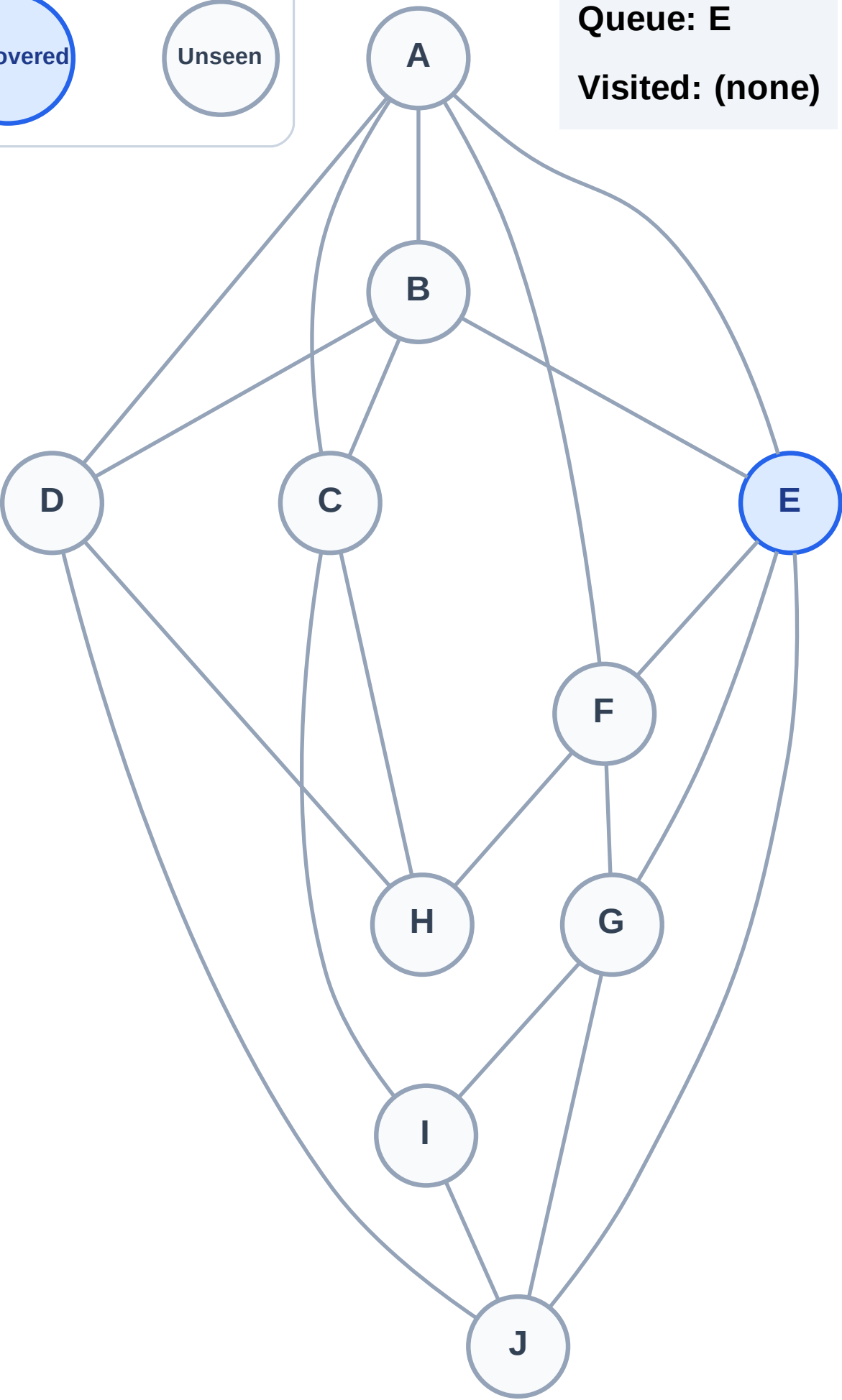
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

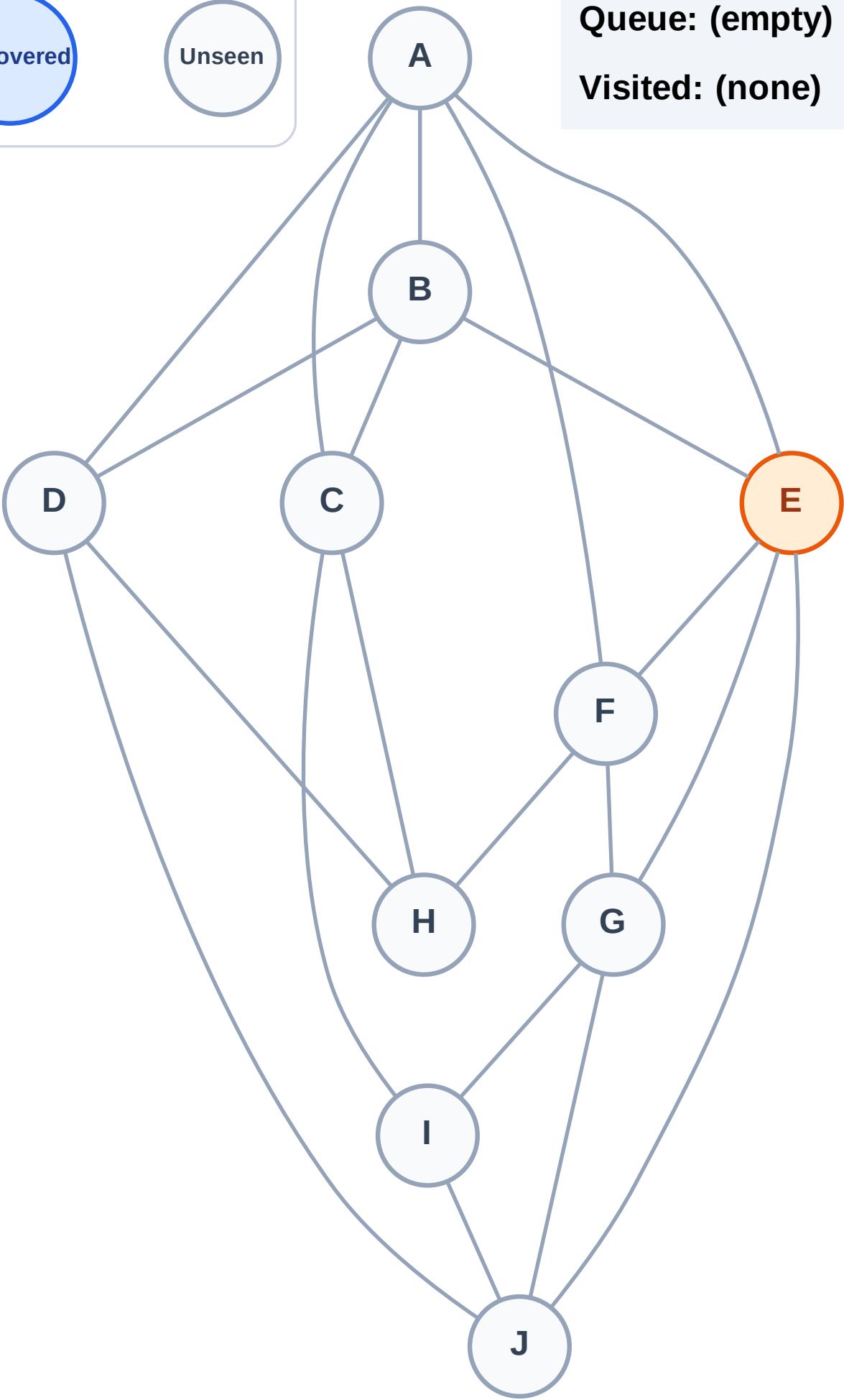
Step 2: Process node E

Legend



Queue: (empty)

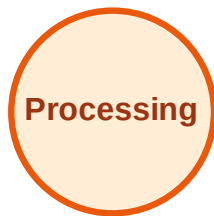
Visited: (none)



BFS Traversal

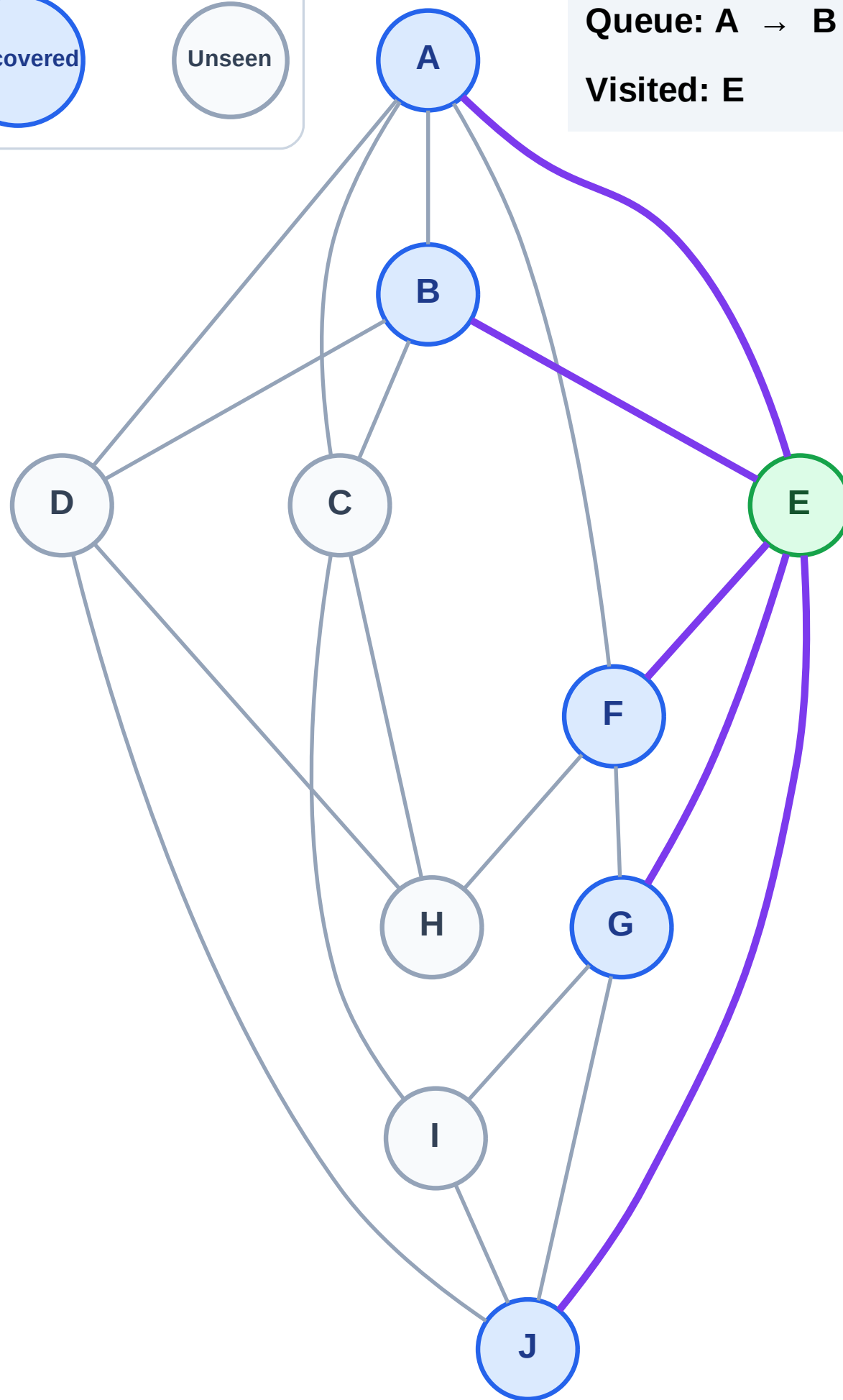
Step 3: Discover A, B, F, G, J from E

Legend



Queue: A → B → F → G → J

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

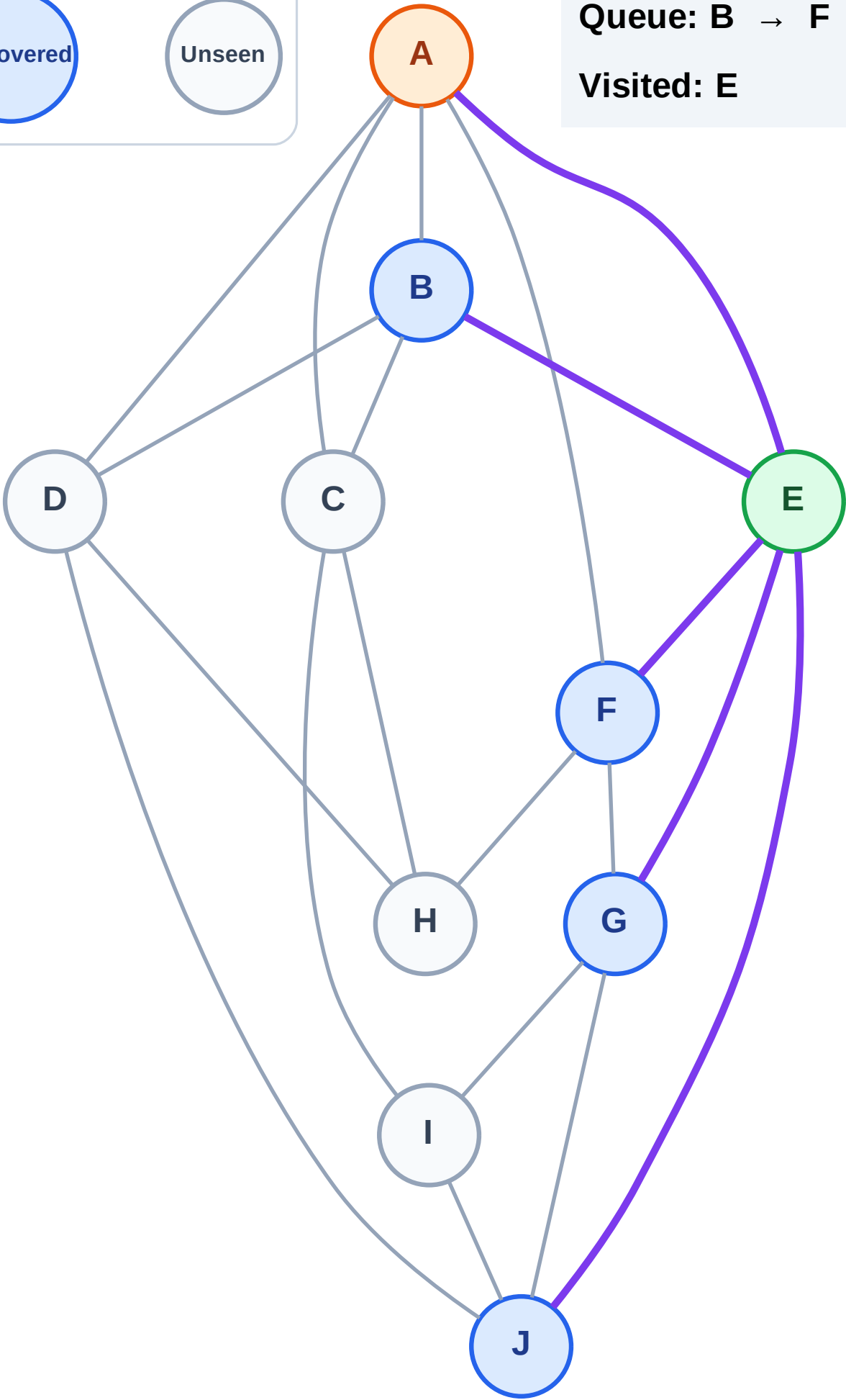
Processing

Discovered

Unseen

Queue: B → F → G → J

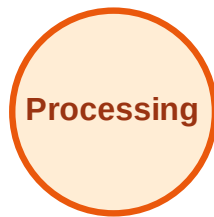
Visited: E



BFS Traversal

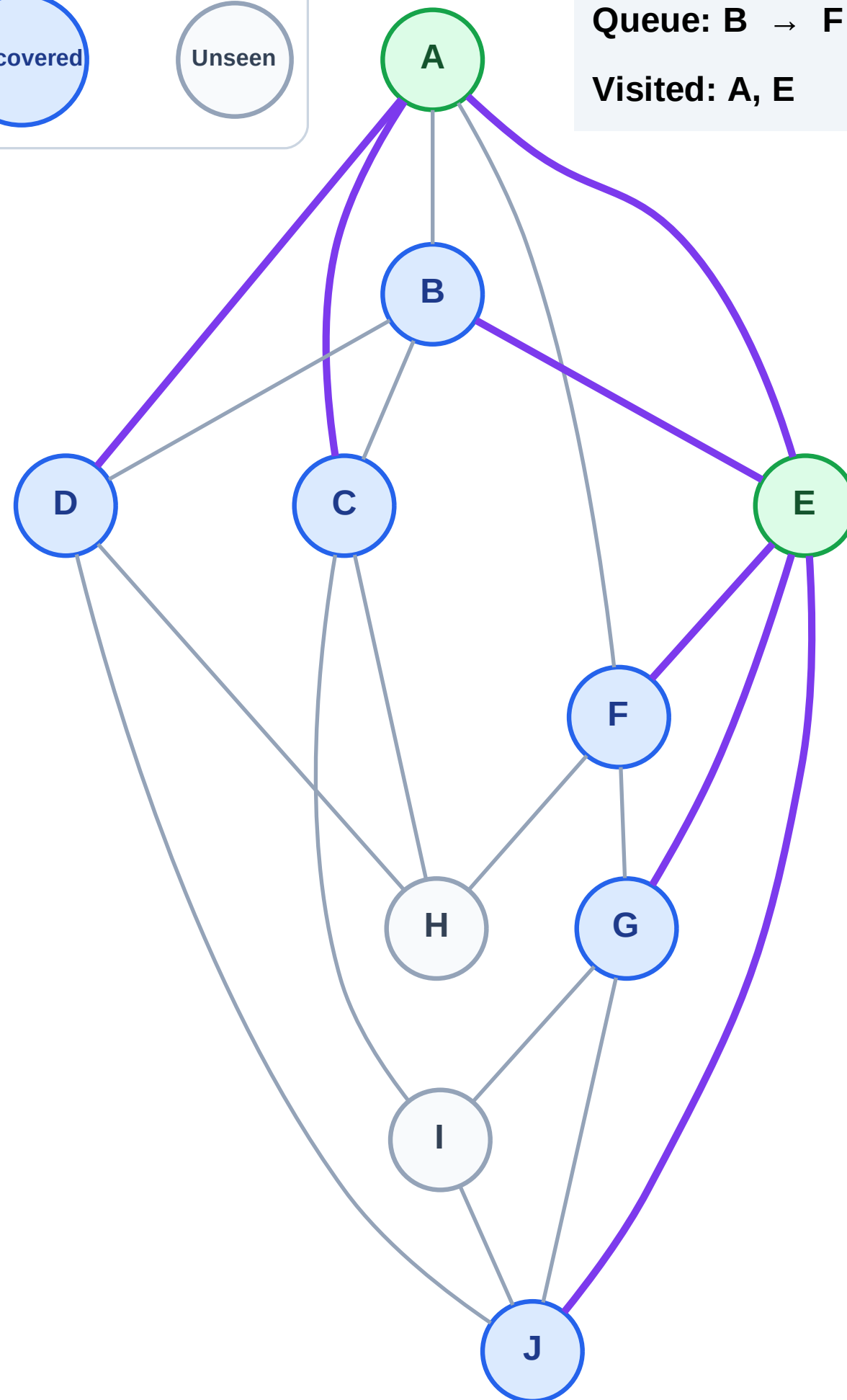
Step 5: Discover C, D from A

Legend



Queue: B → F → G → J → C → D

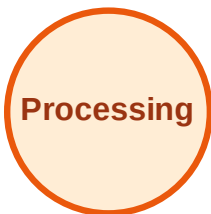
Visited: A, E



BFS Traversal

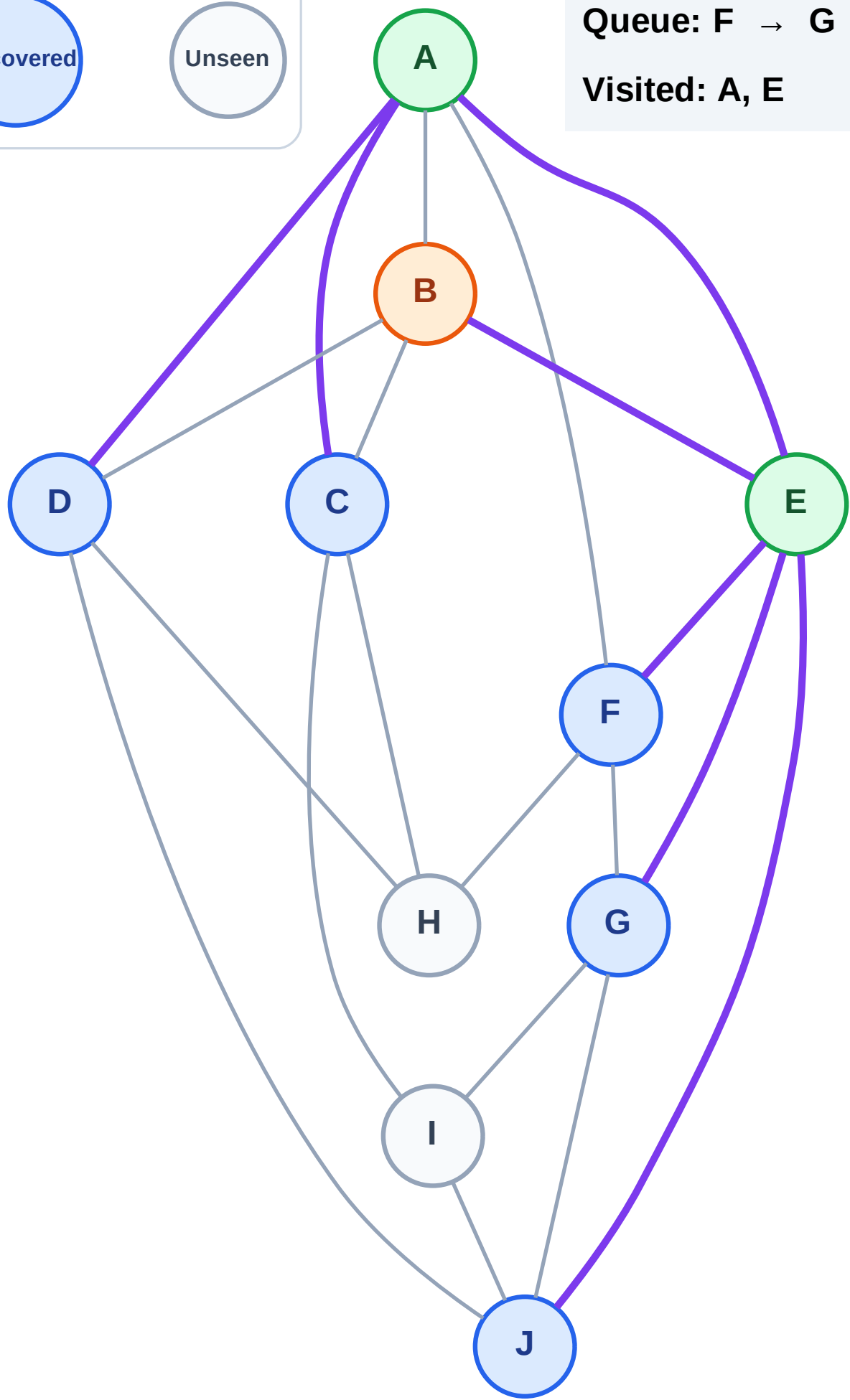
Step 6: Process node B

Legend



Queue: F → G → J → C → D

Visited: A, E



BFS Traversal

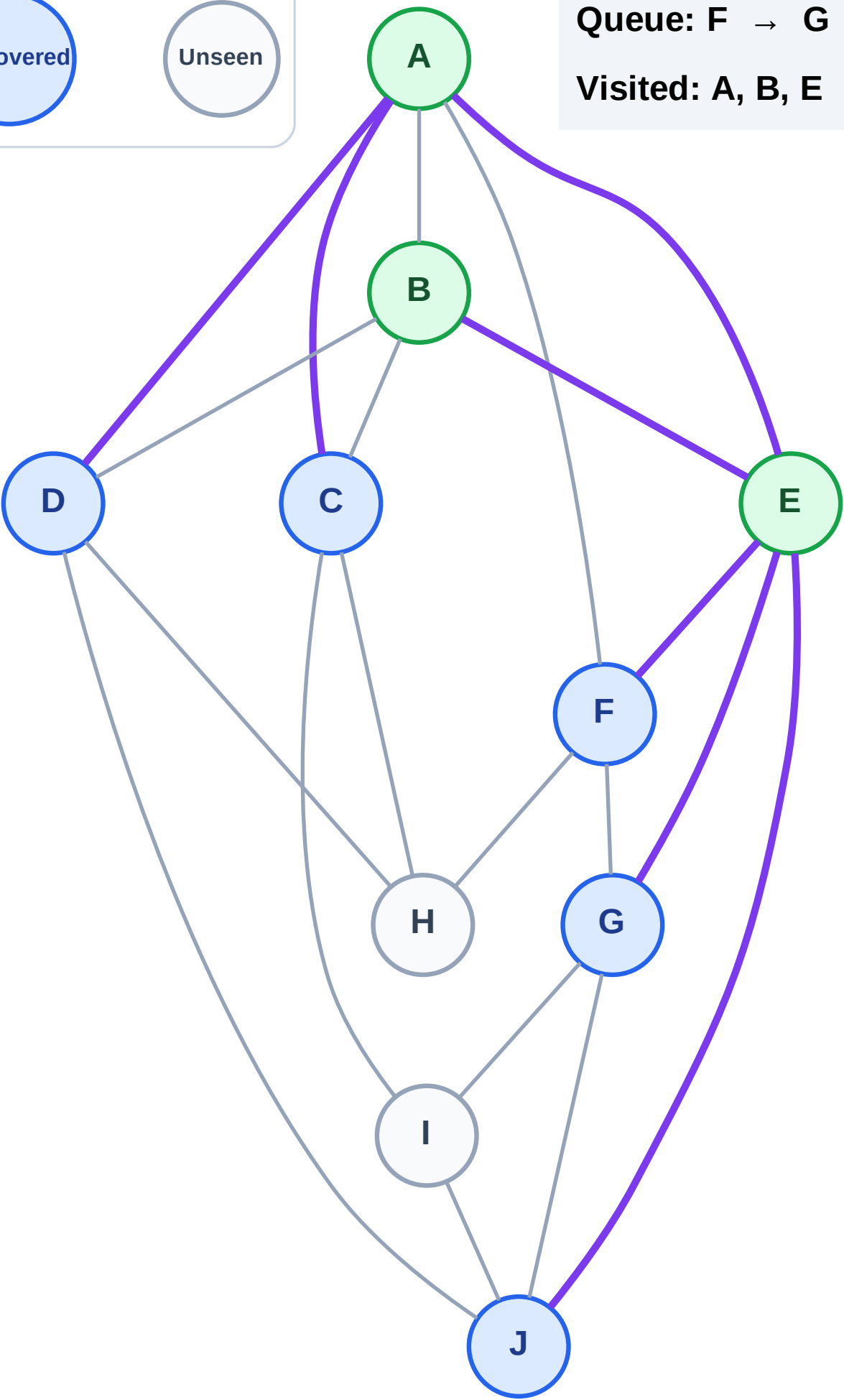
Step 7: No new neighbors from B

Legend



Queue: F → G → J → C → D

Visited: A, B, E



BFS Traversal

Step 8: Process node F

Legend

Complete

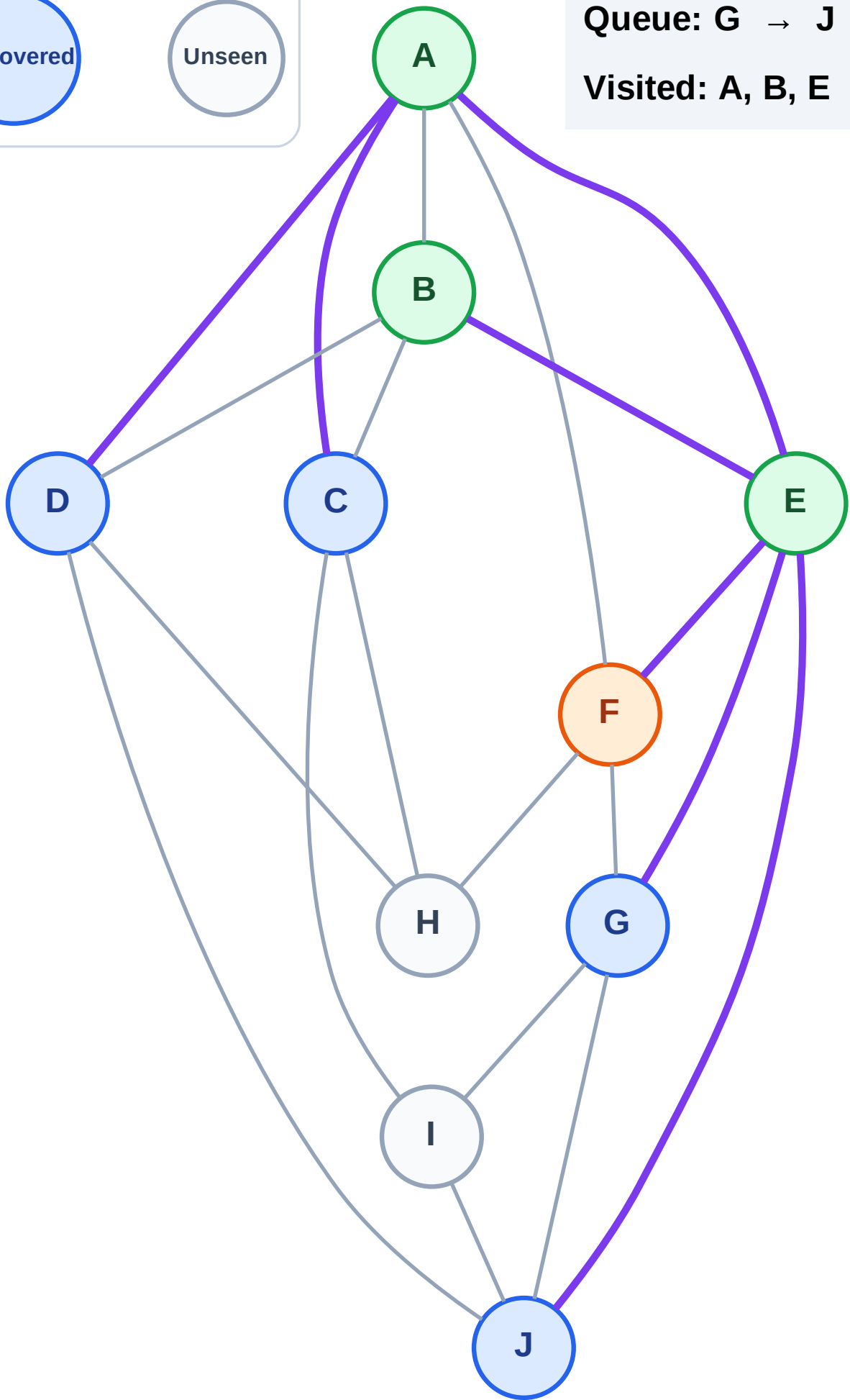
Processing

Discovered

Unseen

Queue: G → J → C → D

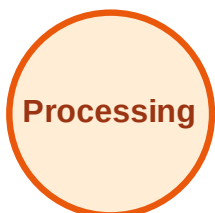
Visited: A, B, E



BFS Traversal

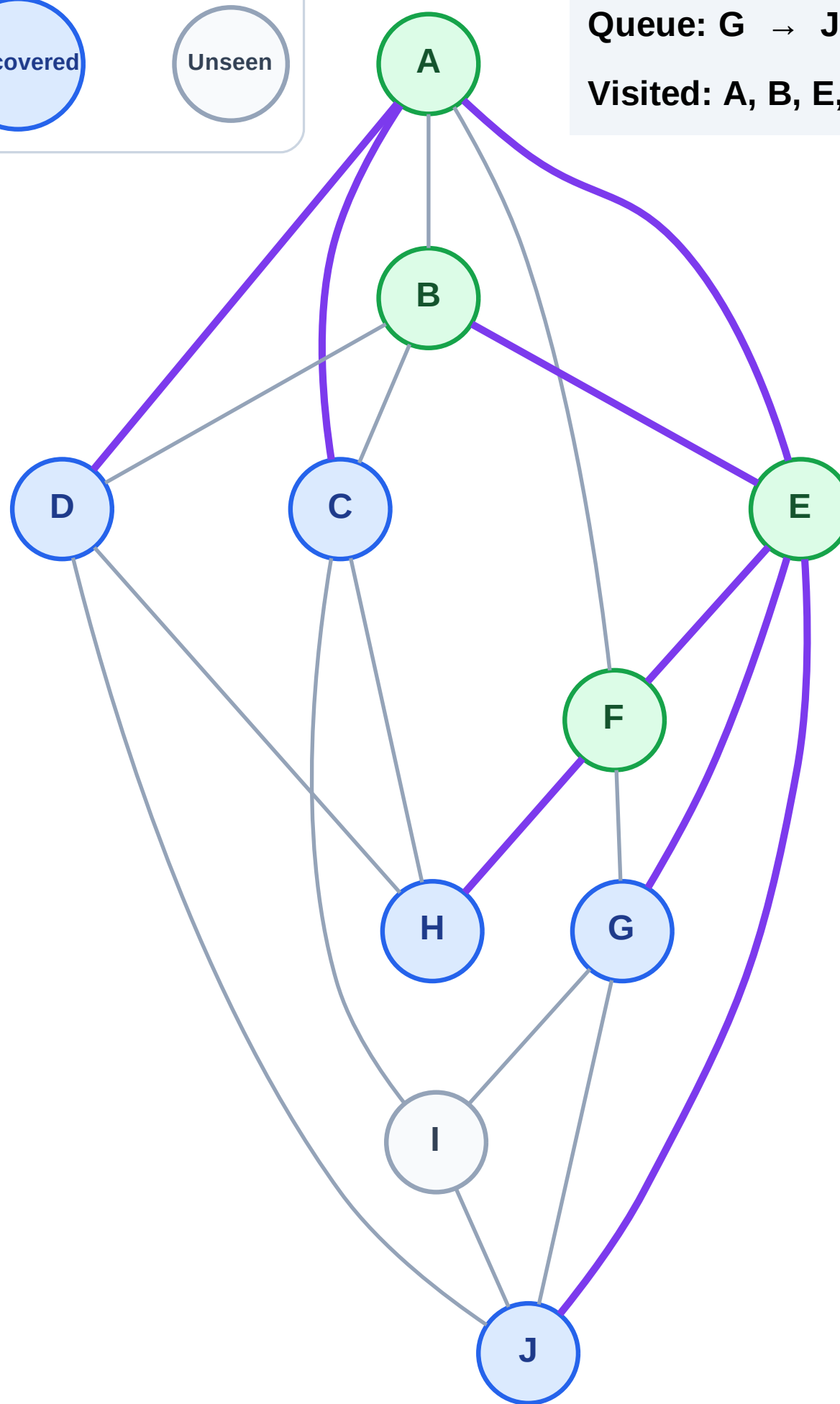
Step 9: Discover H from F

Legend



Queue: G → J → C → D → H

Visited: A, B, E, F



BFS Traversal

Step 10: Process node G

Legend

Complete

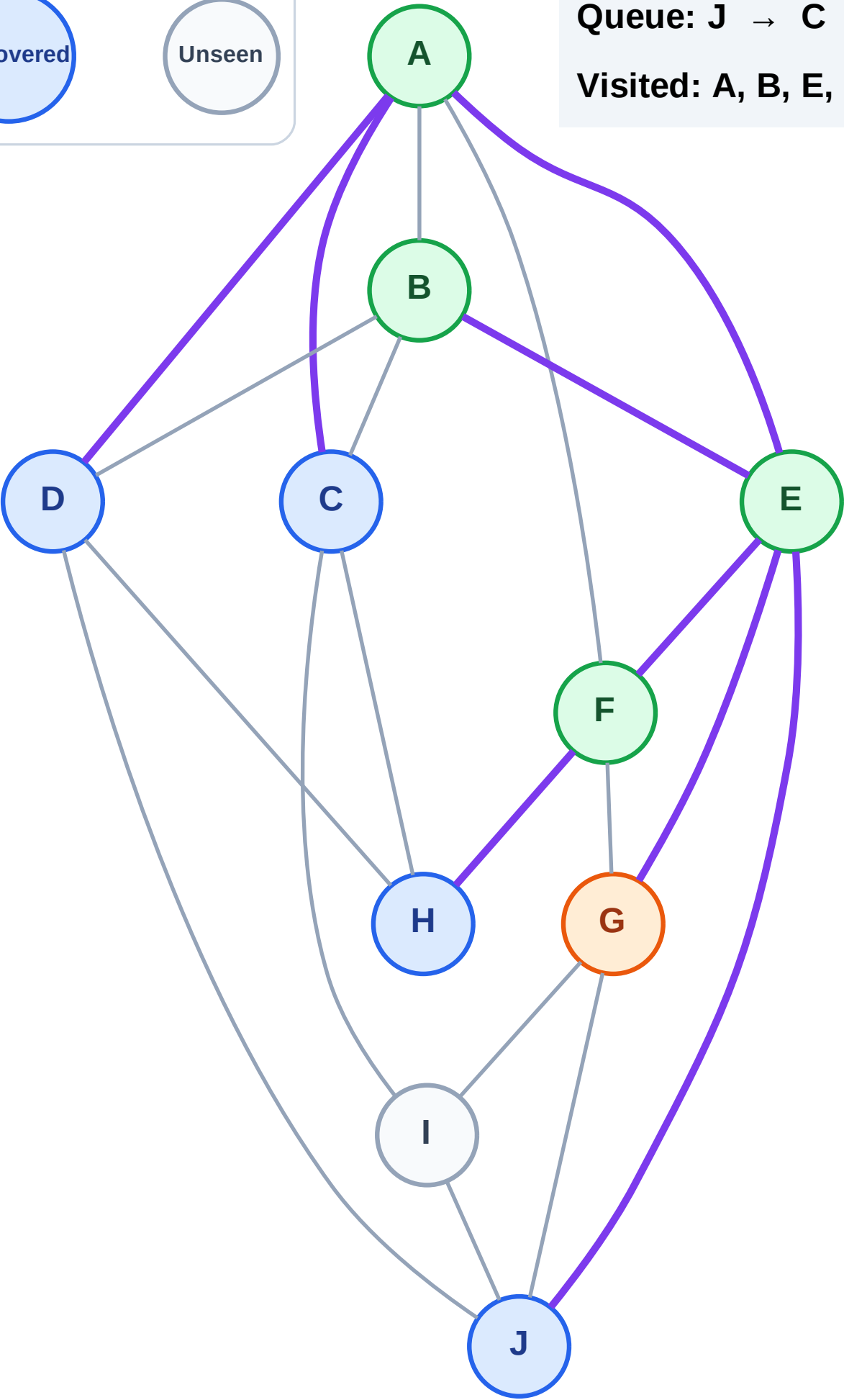
Processing

Discovered

Unseen

Queue: J → C → D → H

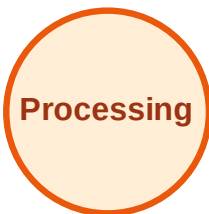
Visited: A, B, E, F



BFS Traversal

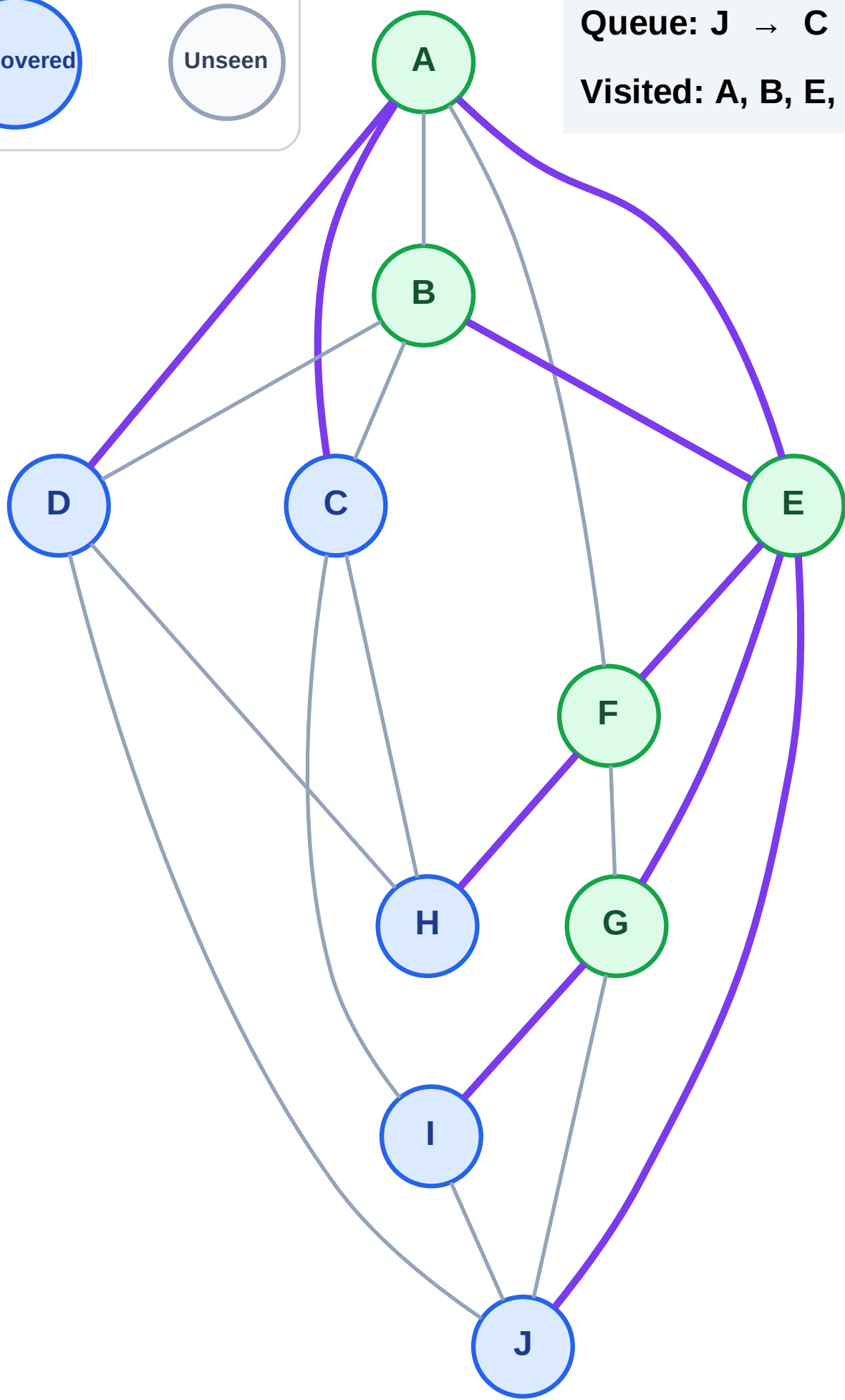
Step 11: Discover I from G

Legend



Queue: J → C → D → H → I

Visited: A, B, E, F, G



BFS Traversal

Step 12: Process node J

Legend

Complete

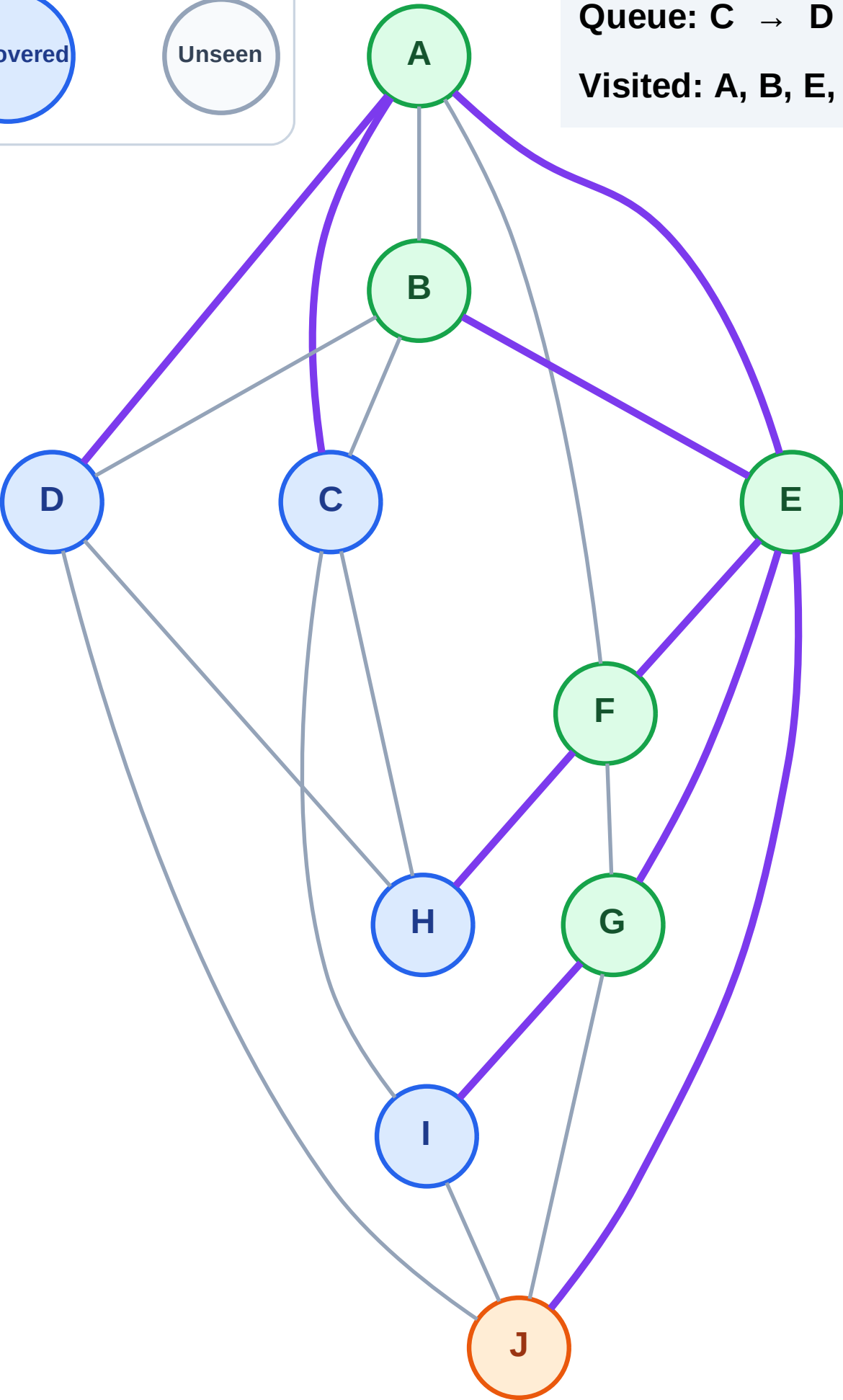
Processing

Discovered

Unseen

Queue: C → D → H → I

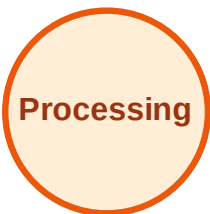
Visited: A, B, E, F, G



BFS Traversal

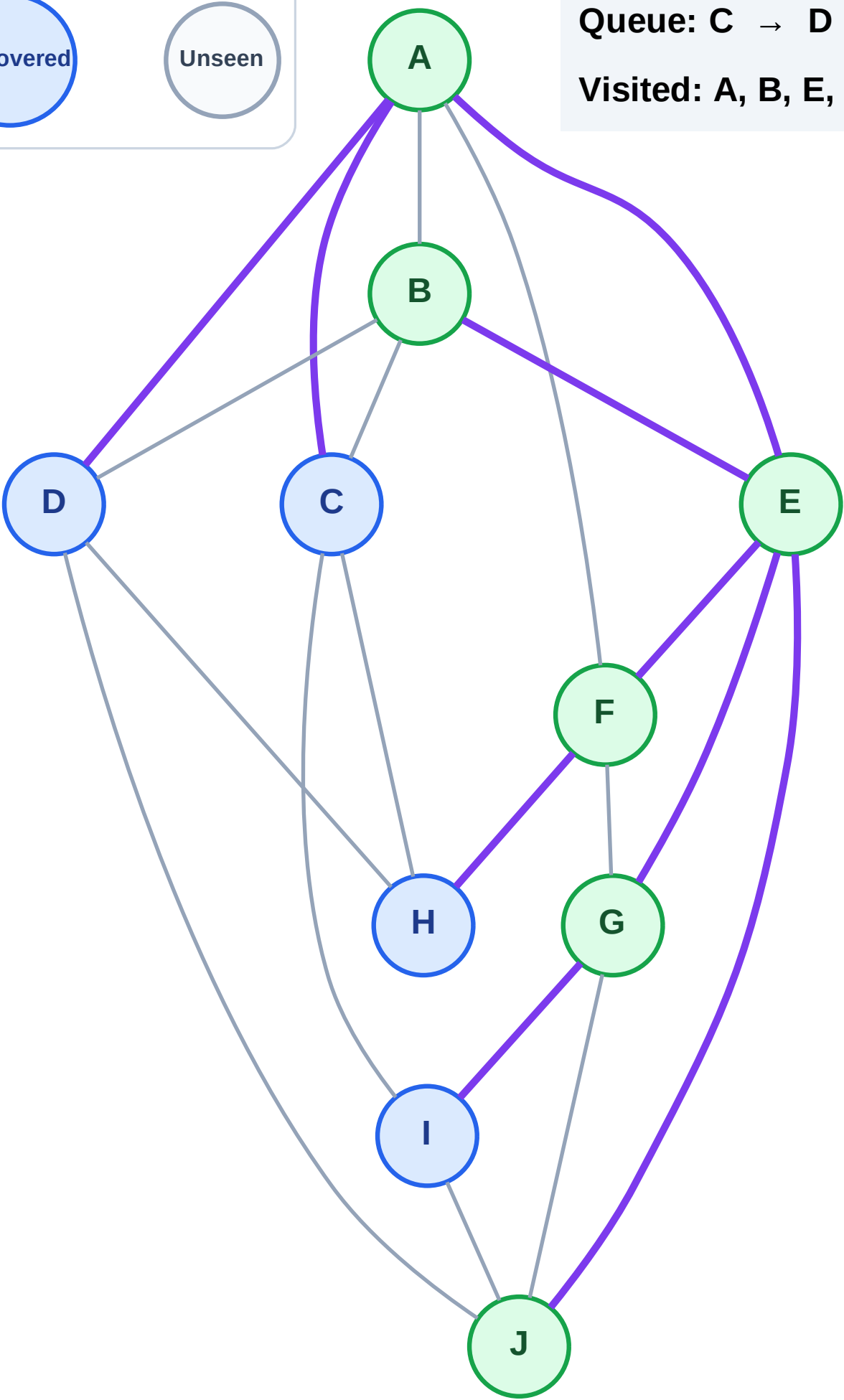
Step 13: No new neighbors from J

Legend



Queue: C → D → H → I

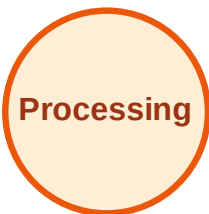
Visited: A, B, E, F, G, J



BFS Traversal

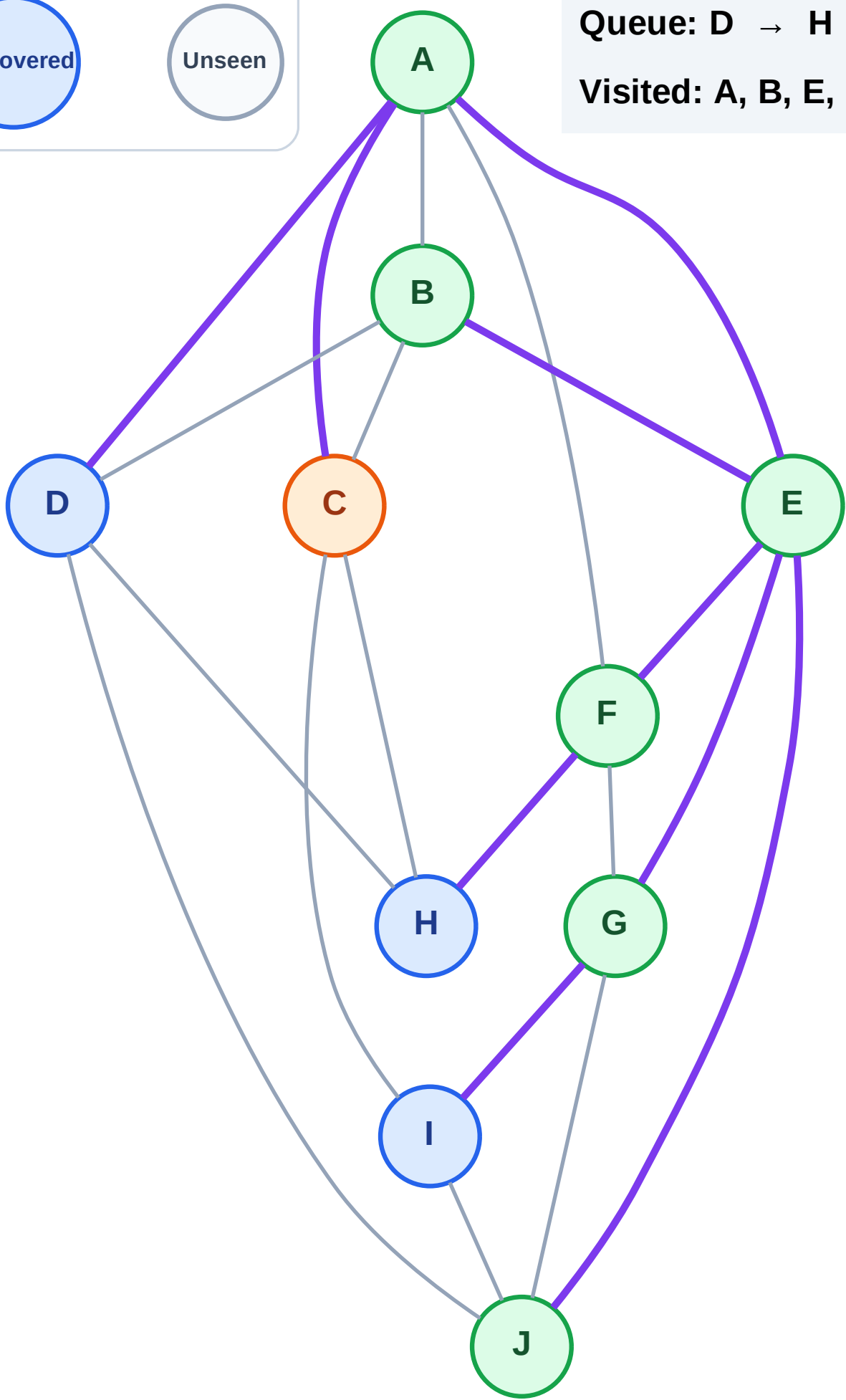
Step 14: Process node C

Legend



Queue: D → H → I

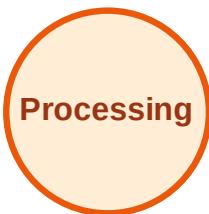
Visited: A, B, E, F, G, J



BFS Traversal

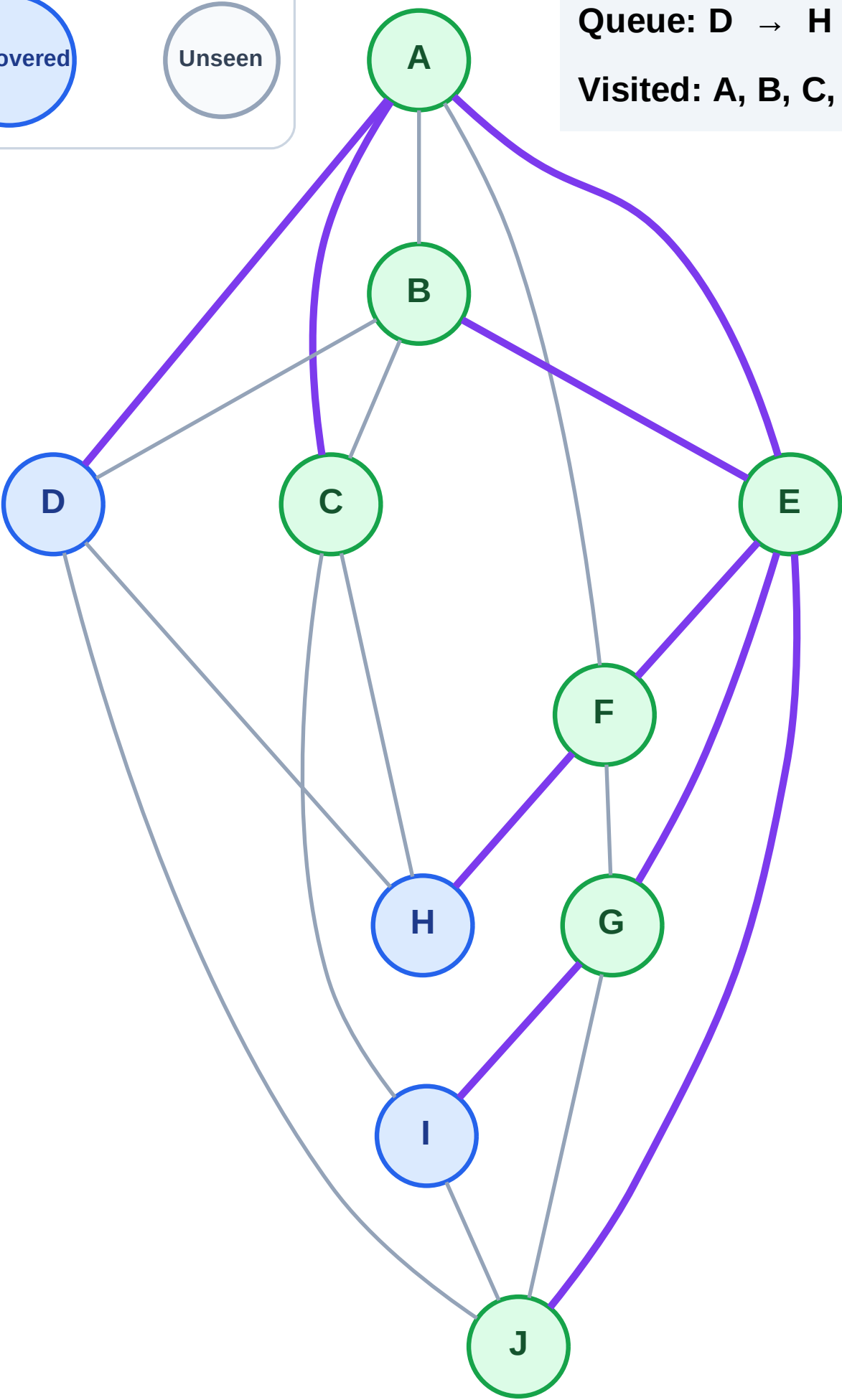
Step 15: No new neighbors from C

Legend



Queue: D → H → I

Visited: A, B, C, E, F, G, J



BFS Traversal

Step 16: Process node D

Legend

Complete

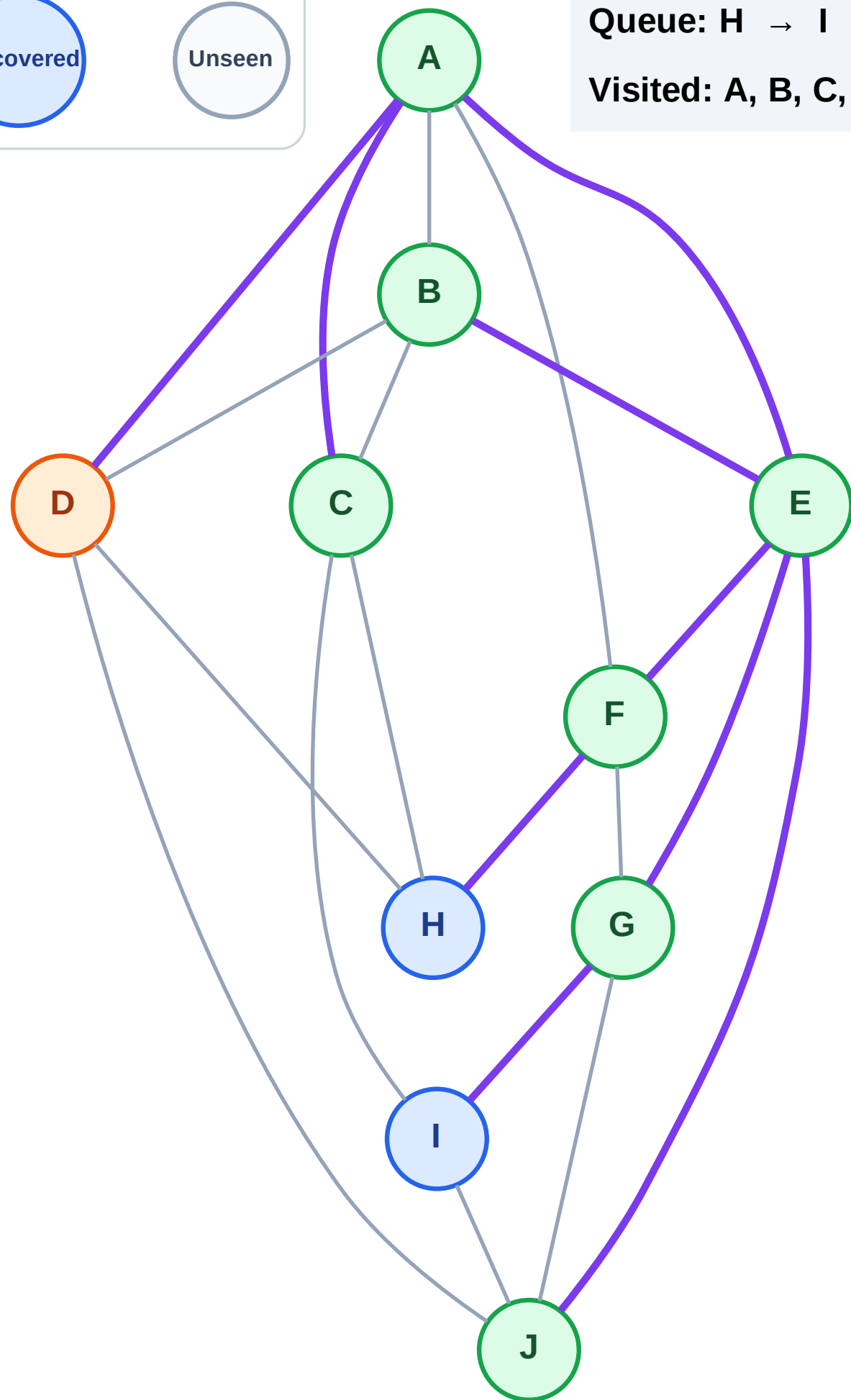
Processing

Discovered

Unseen

Queue: H → I

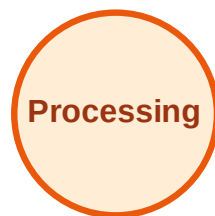
Visited: A, B, C, E, F, G, J



BFS Traversal

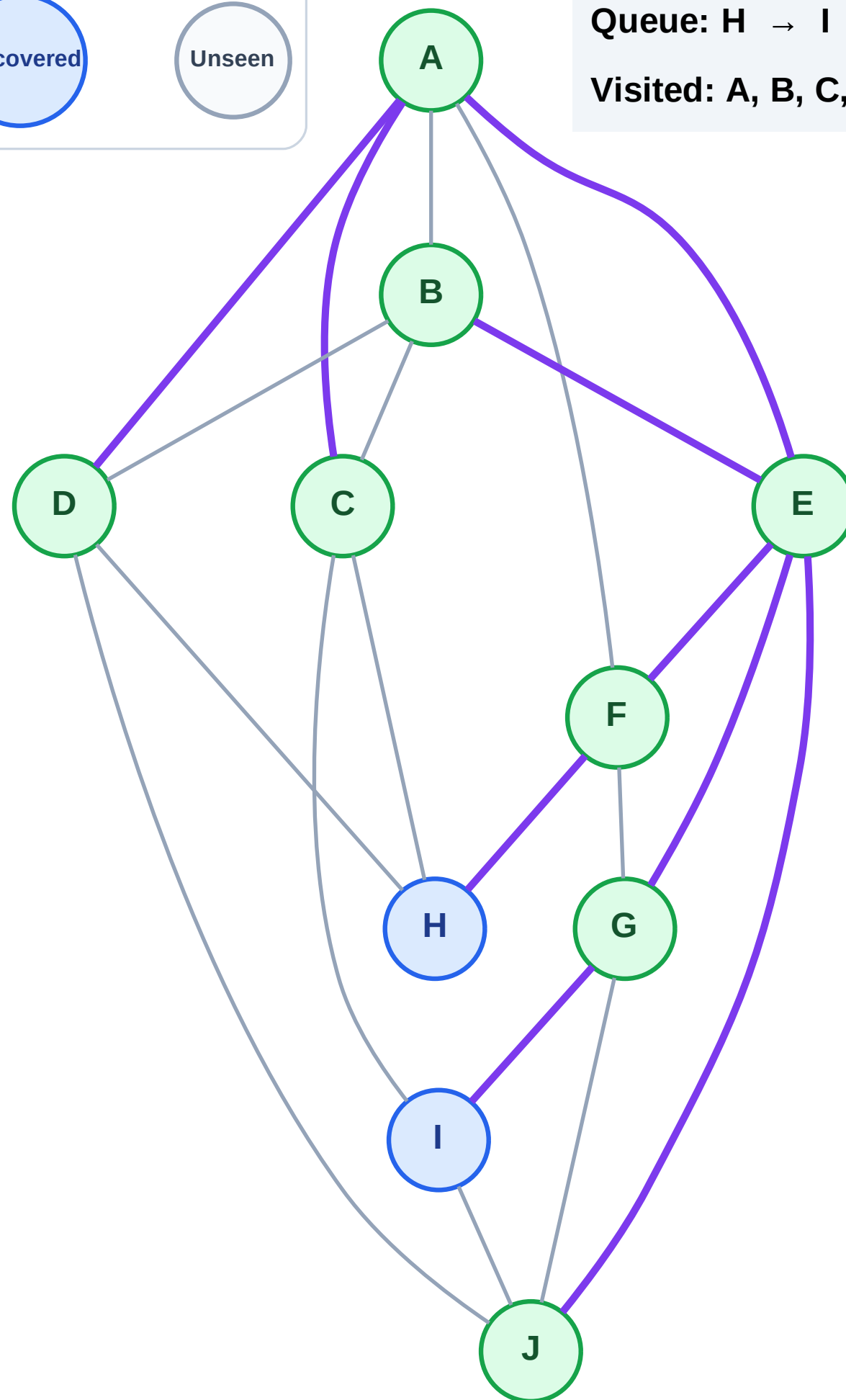
Step 17: No new neighbors from D

Legend



Queue: H → I

Visited: A, B, C, D, E, F, G, J



BFS Traversal

Step 18: Process node H

Legend

Complete

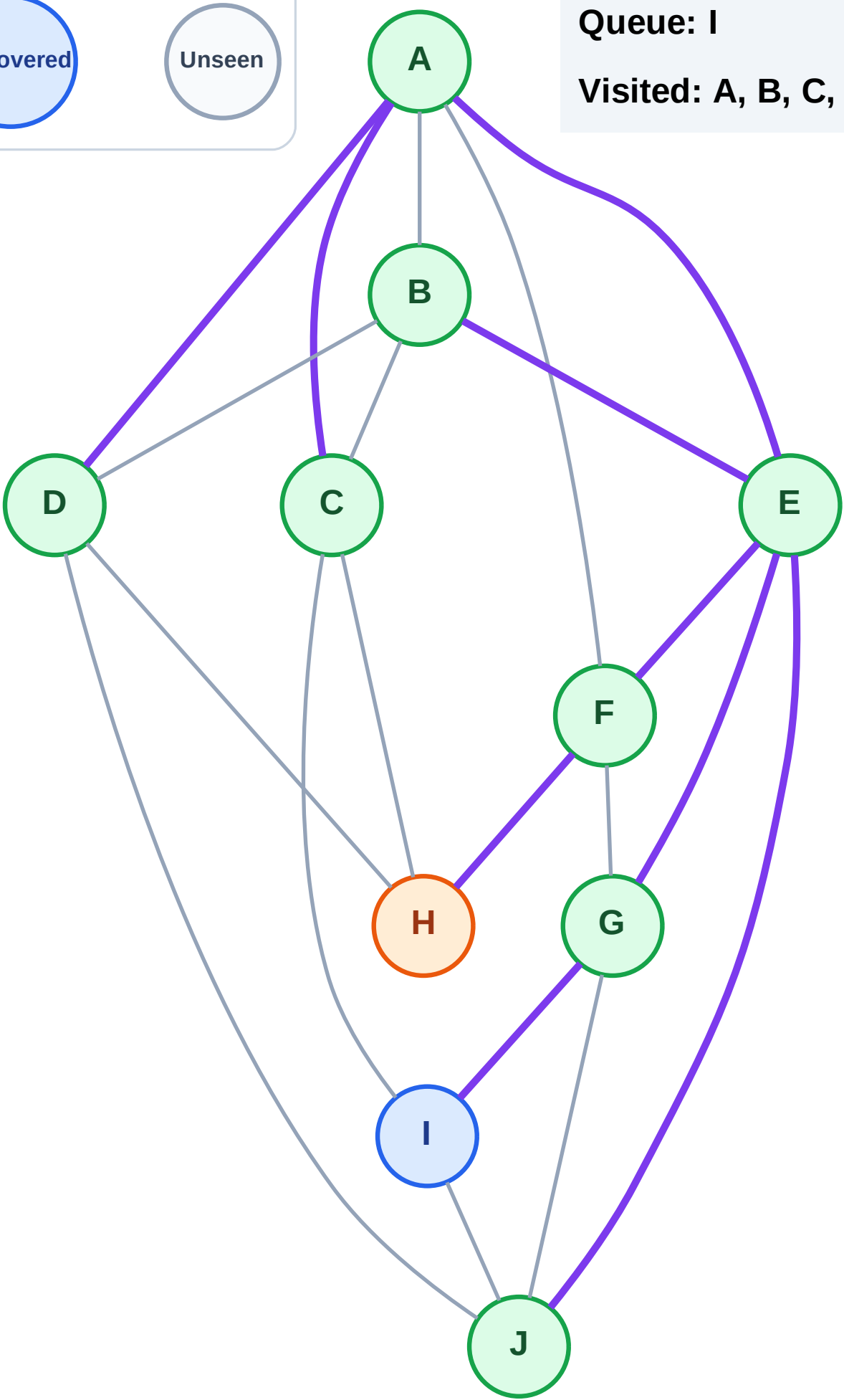
Processing

Discovered

Unseen

Queue: I

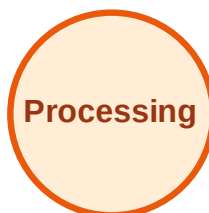
Visited: A, B, C, D, E, F, G, J



BFS Traversal

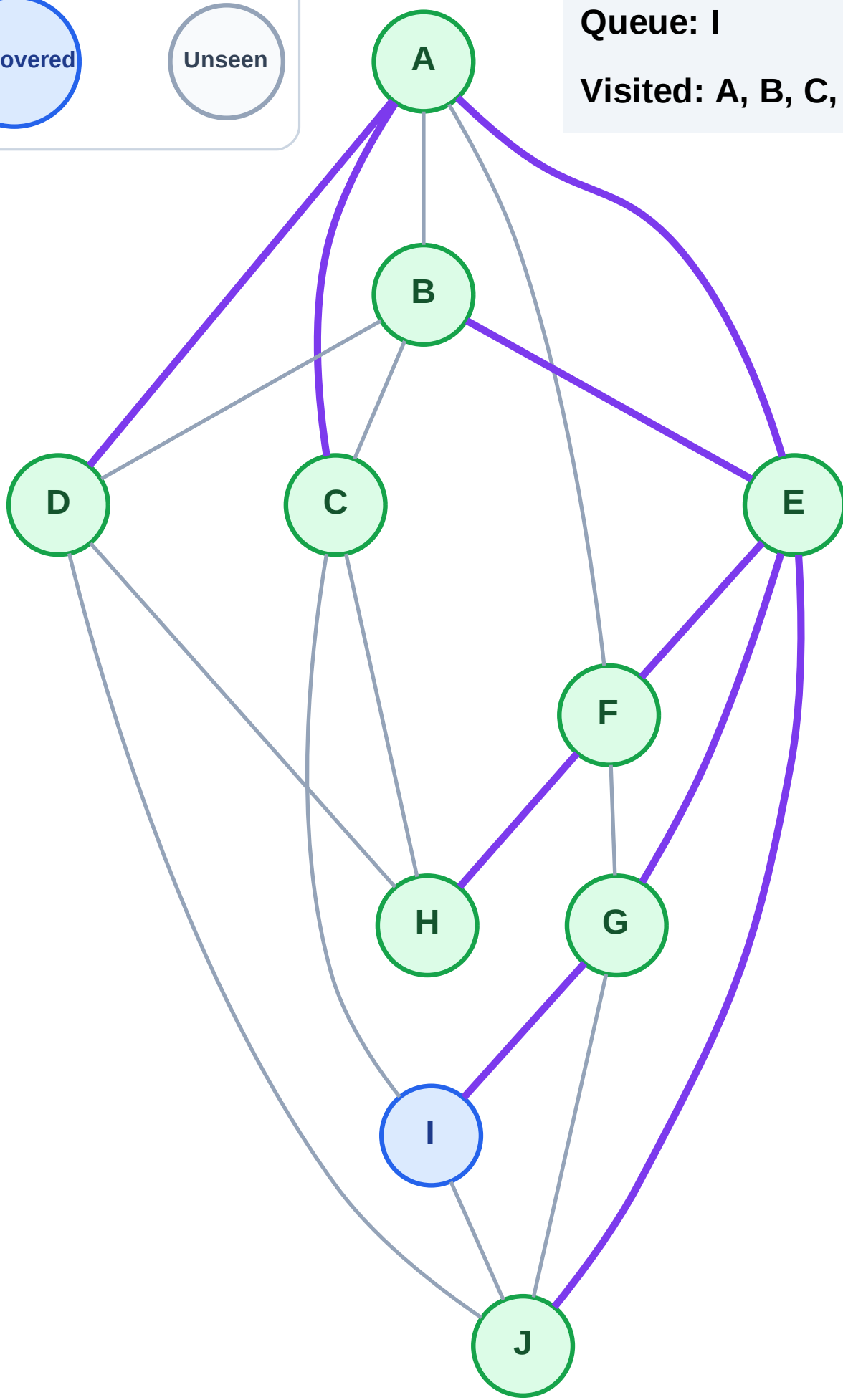
Step 19: No new neighbors from H

Legend



Queue: I

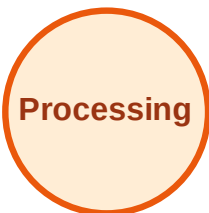
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

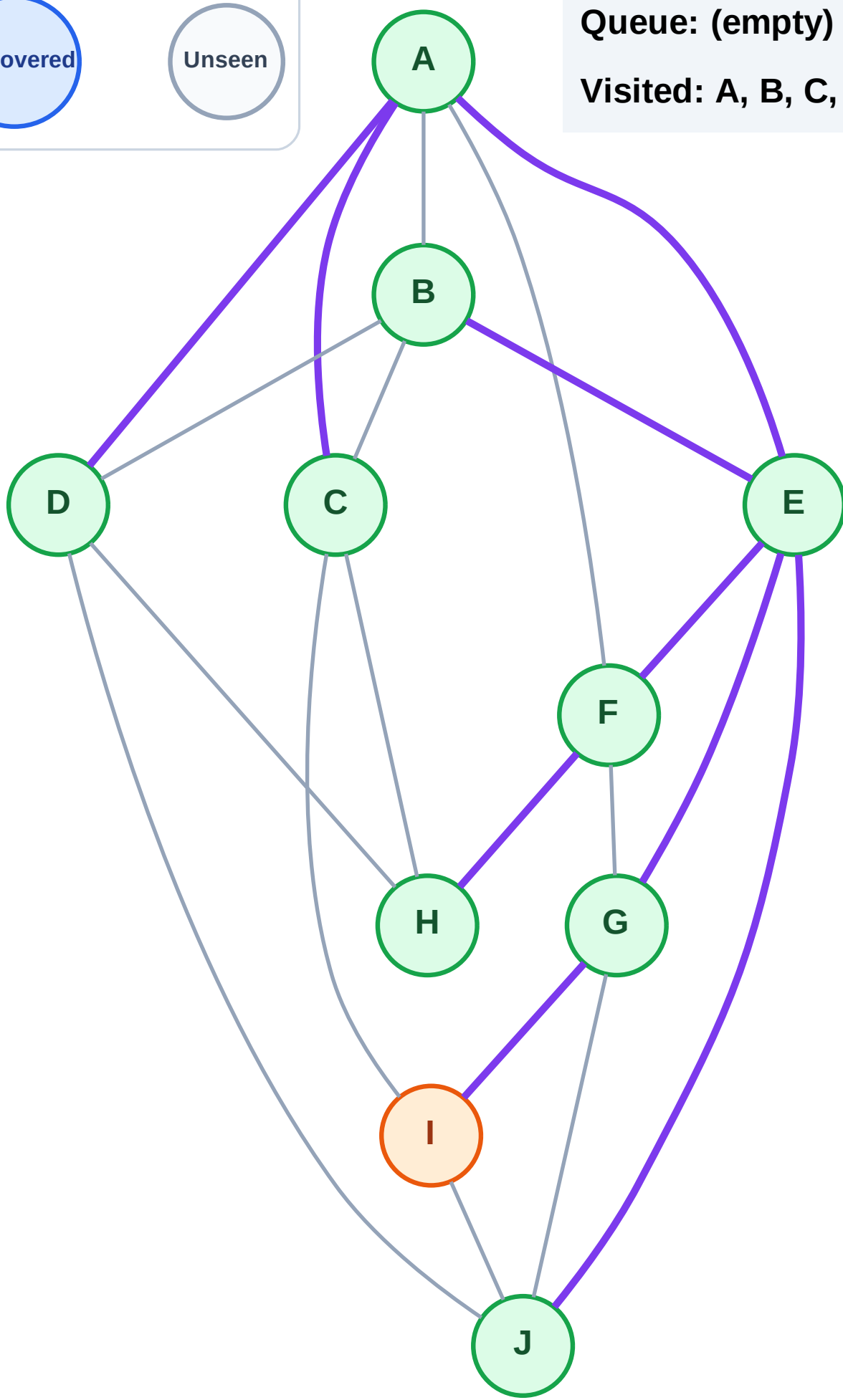
Step 20: Process node I

Legend



Queue: (empty)

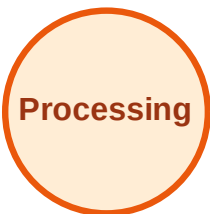
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

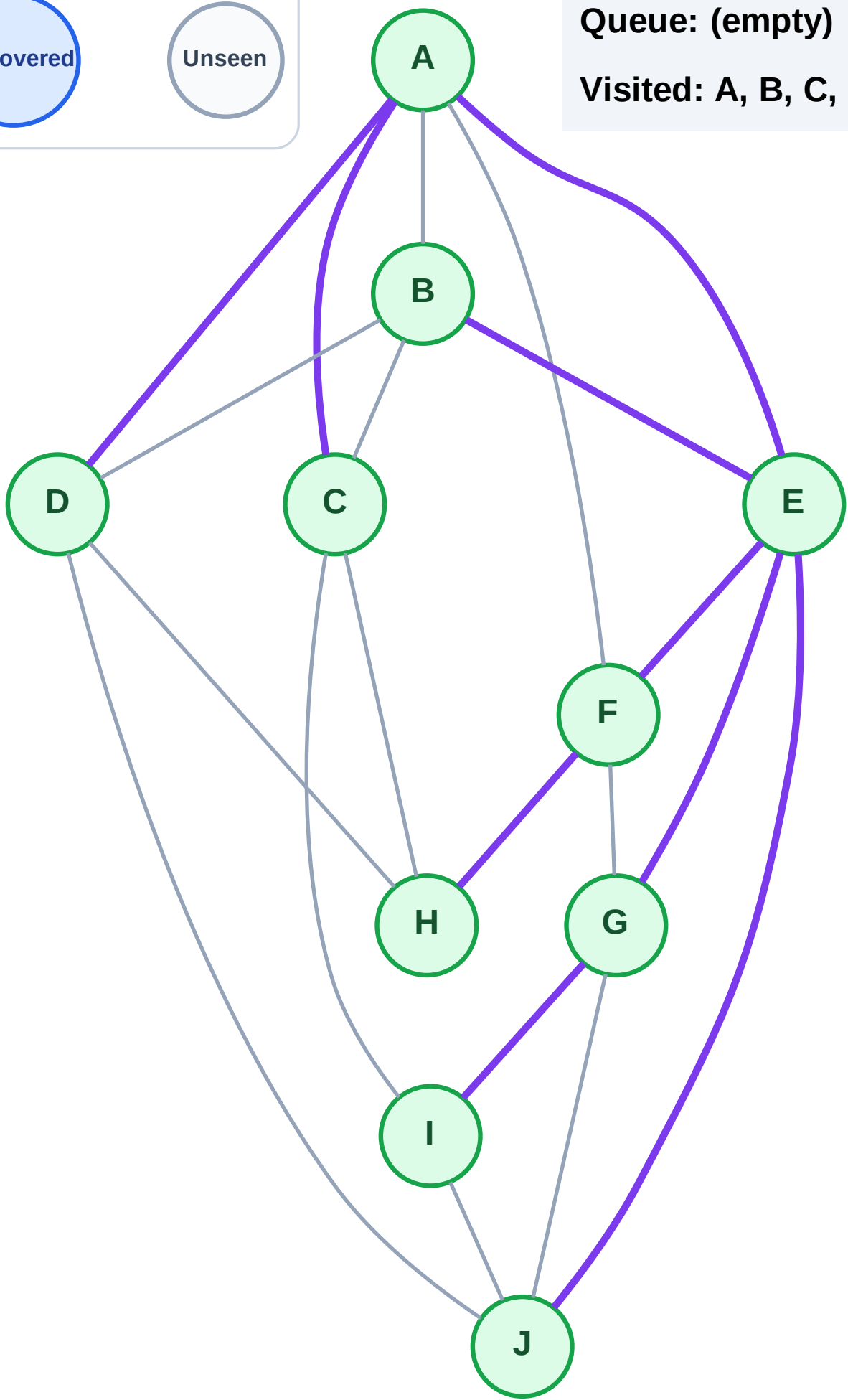
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

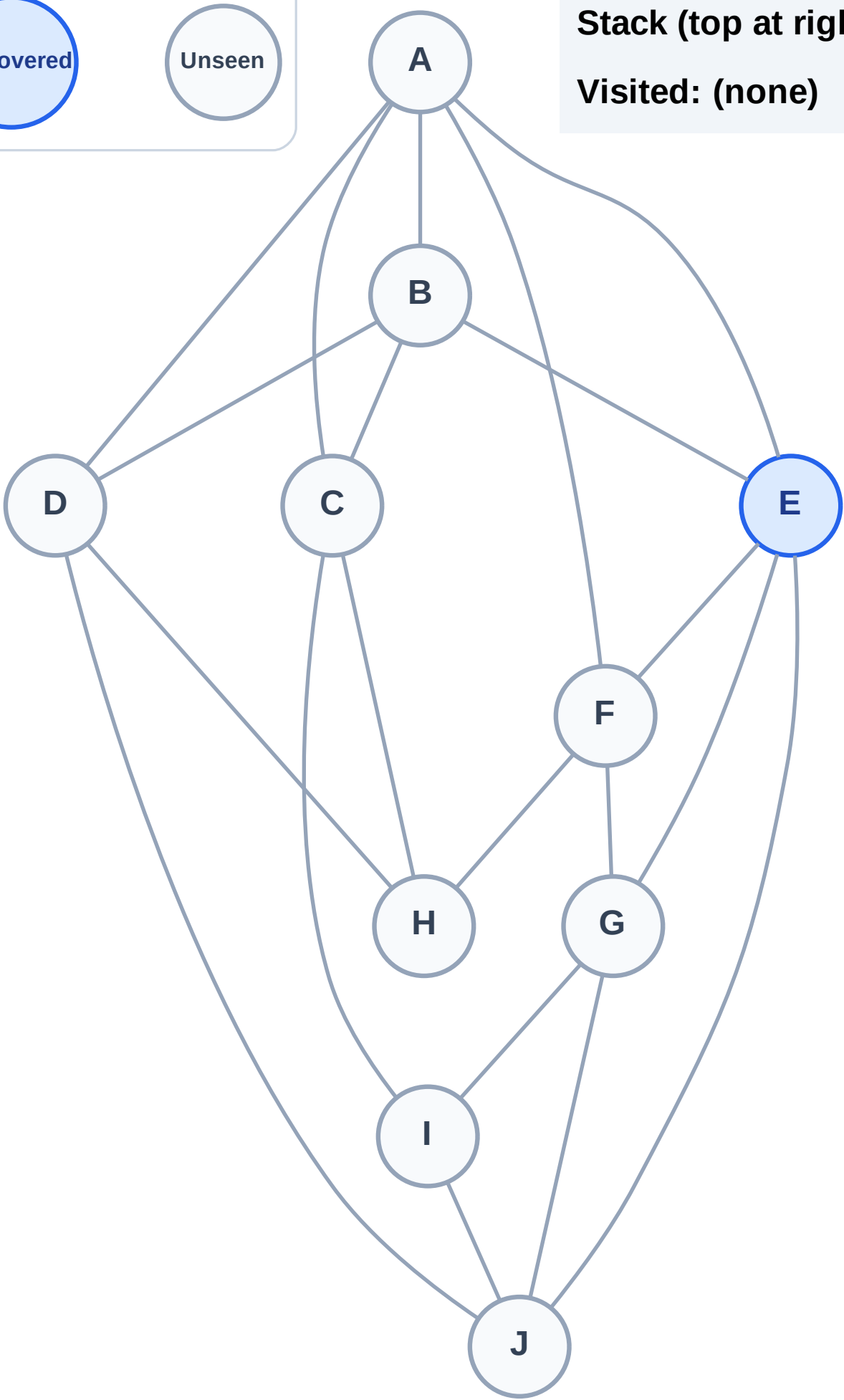
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

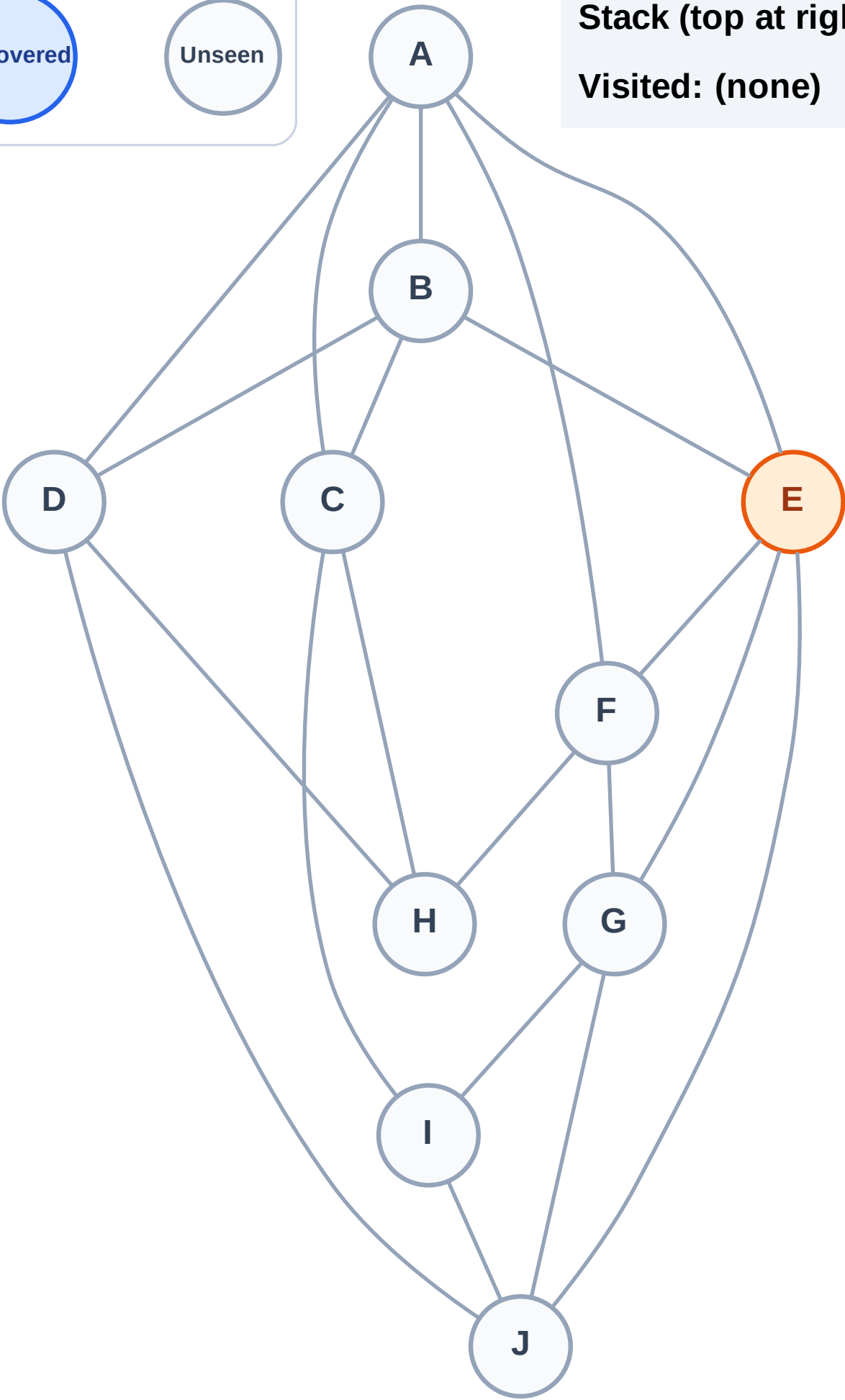
Complete

Processing

Discovered

Unseen

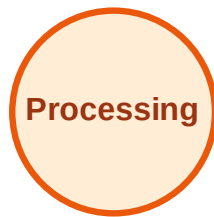
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

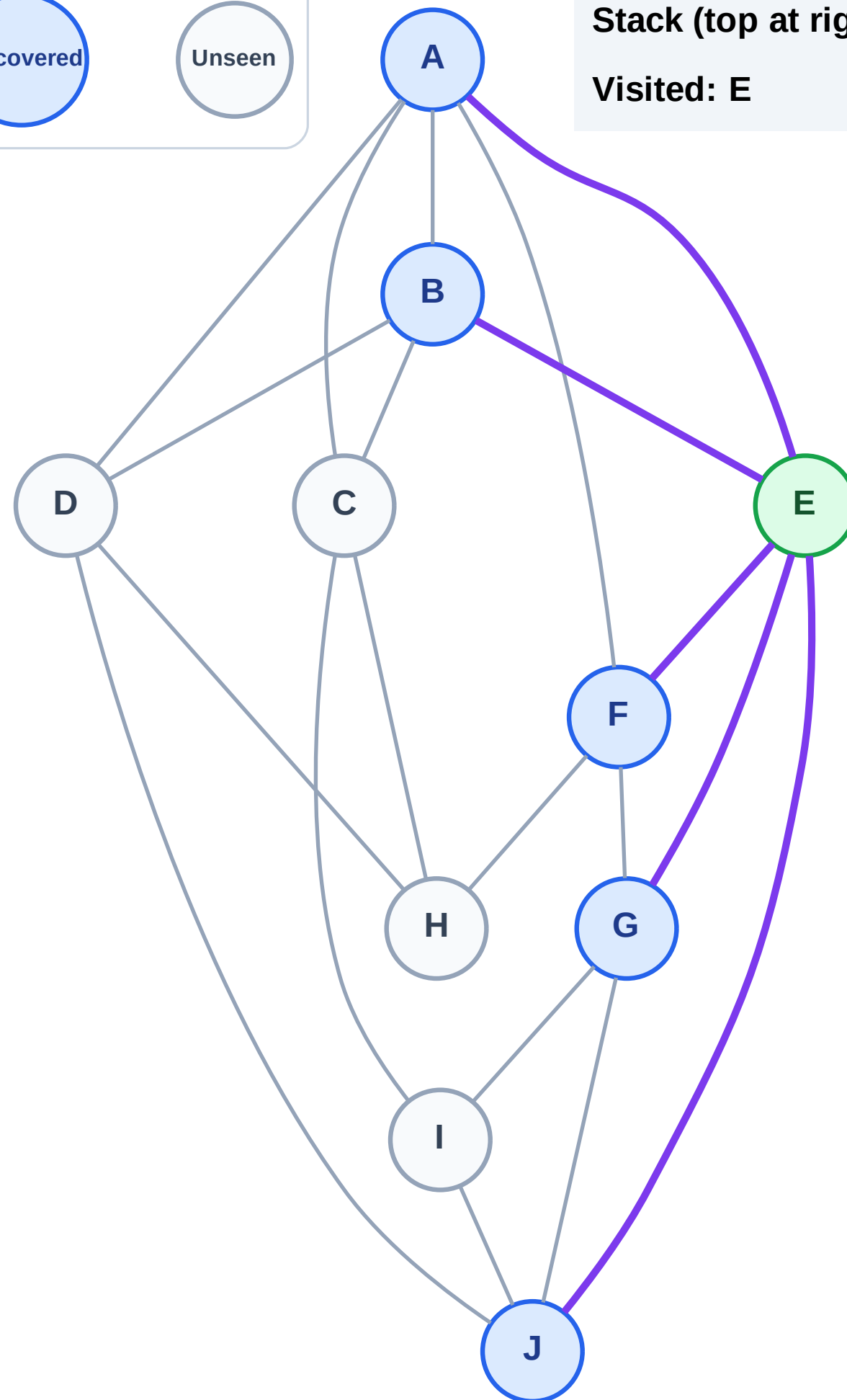
Step 3: Discover J, G, F, B, A from E

Legend



Stack (top at right): J | G | F | B | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

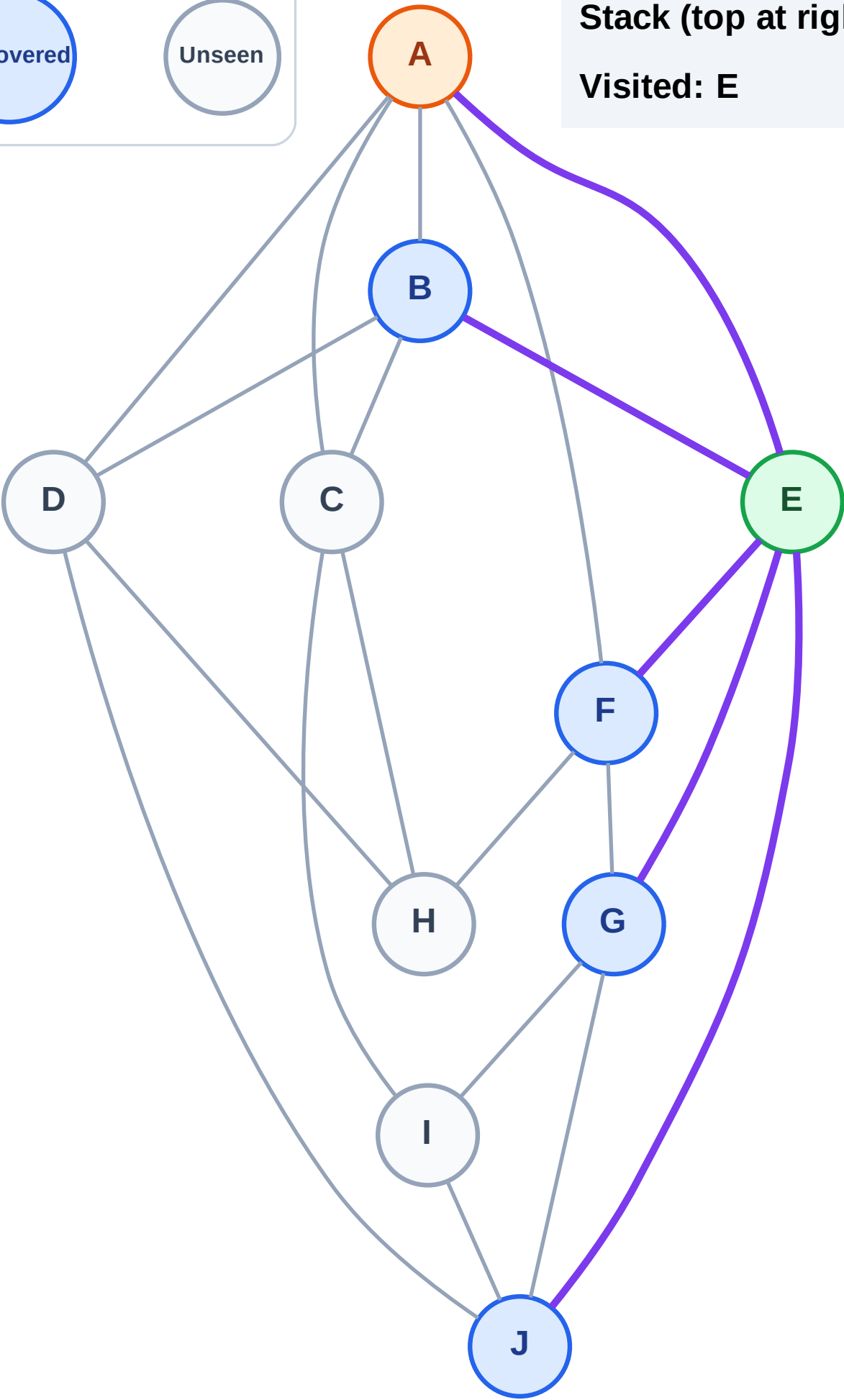
Processing

Discovered

Unseen

Stack (top at right): J | G | F | B

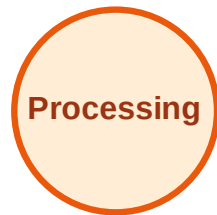
Visited: E



DFS Traversal

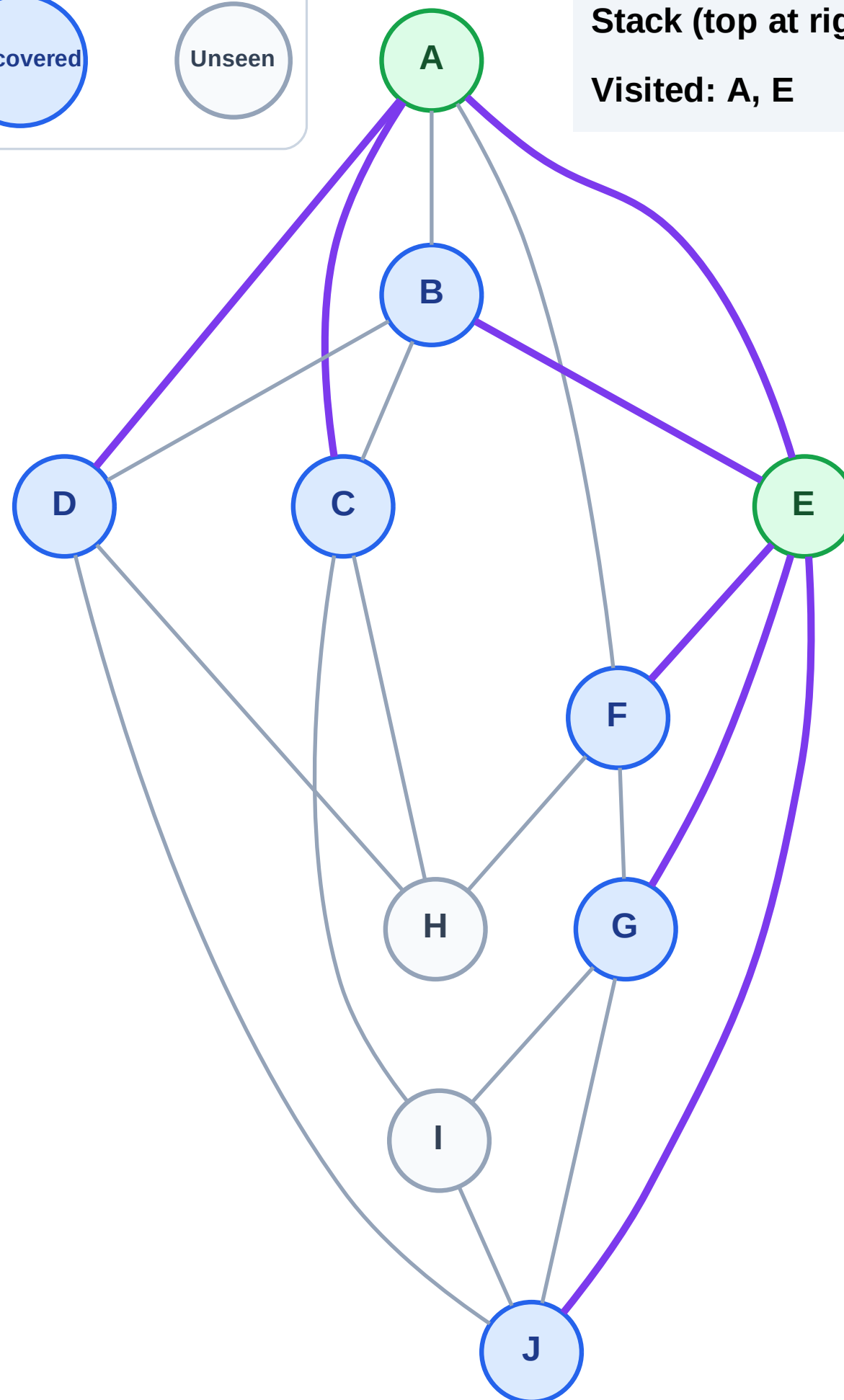
Step 5: Discover D, C from A

Legend



Stack (top at right): J | G | F | B | D | C

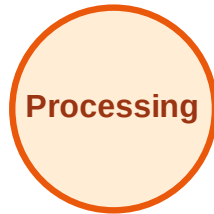
Visited: A, E



DFS Traversal

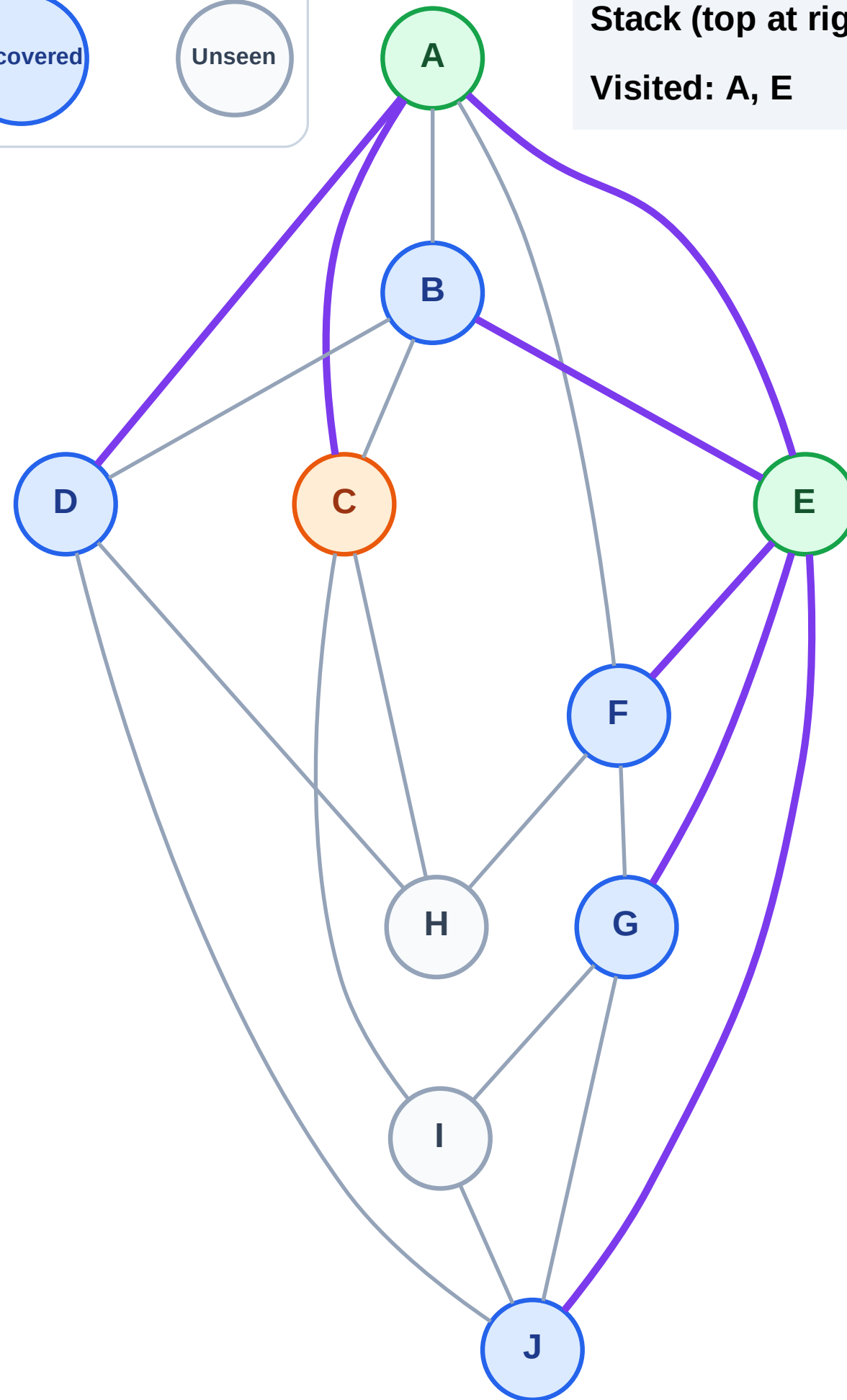
Step 6: Process node C

Legend



Stack (top at right): J | G | F | B | D

Visited: A, E



DFS Traversal

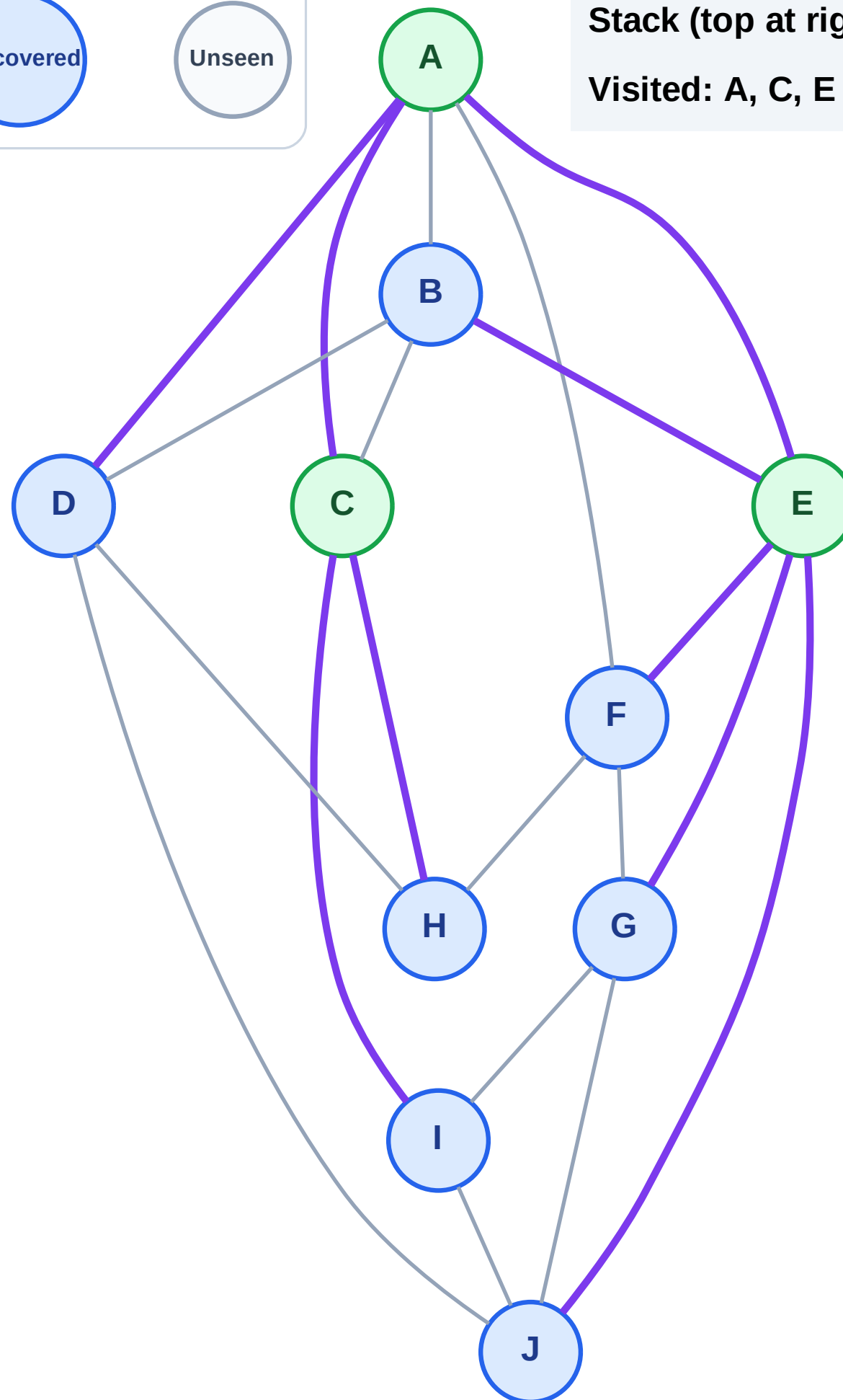
Step 7: Discover I, H from C

Legend



Stack (top at right): J | G | F | B | D | I | H

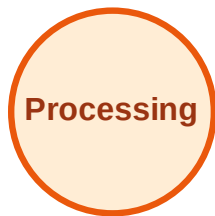
Visited: A, C, E



DFS Traversal

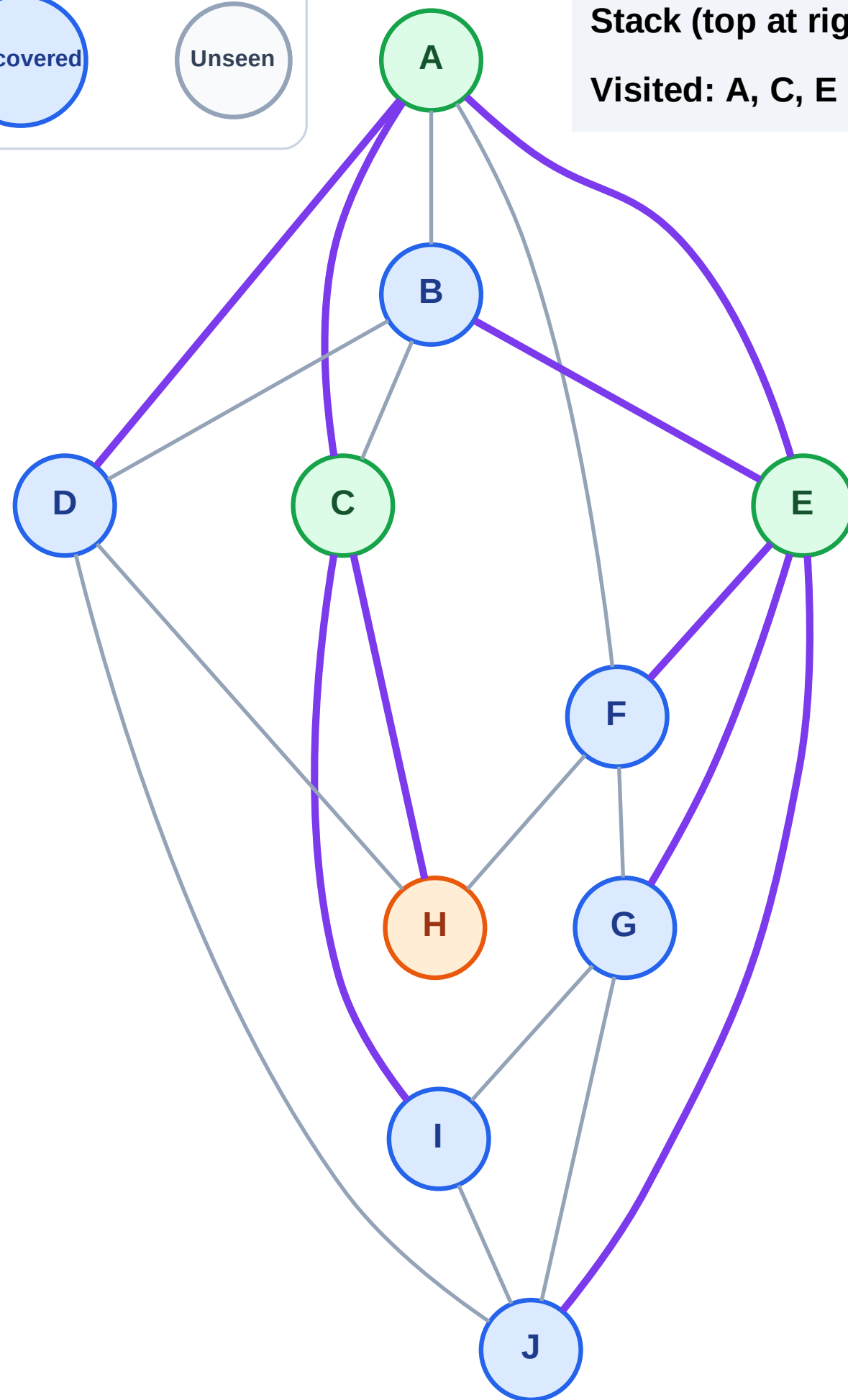
Step 8: Process node H

Legend



Stack (top at right): J | G | F | B | D | I

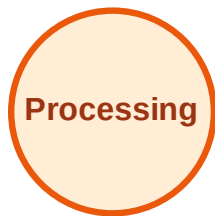
Visited: A, C, E



DFS Traversal

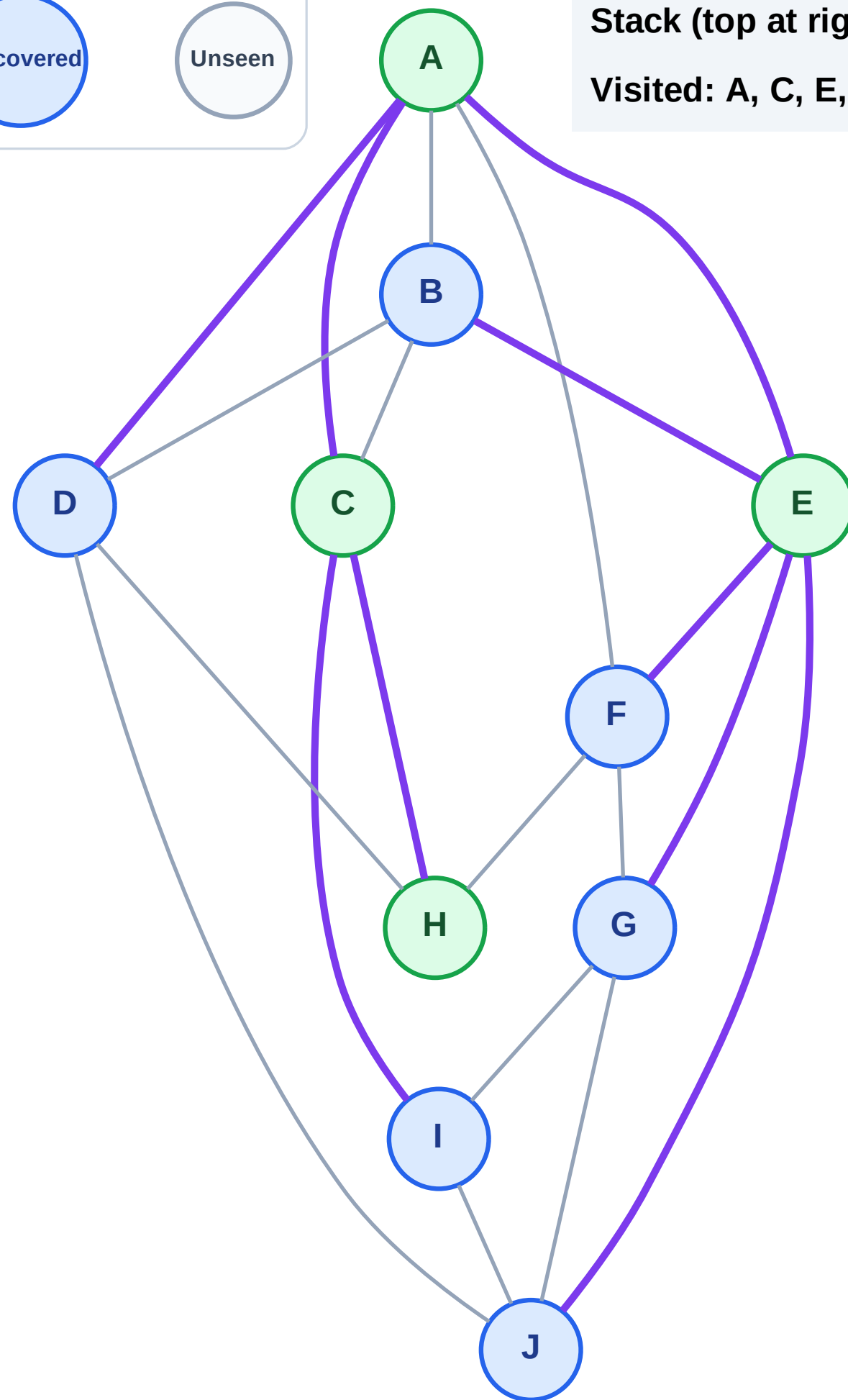
Step 9: No new neighbors from H

Legend



Stack (top at right): J | G | F | B | D | I

Visited: A, C, E, H



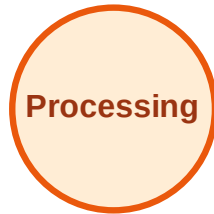
DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



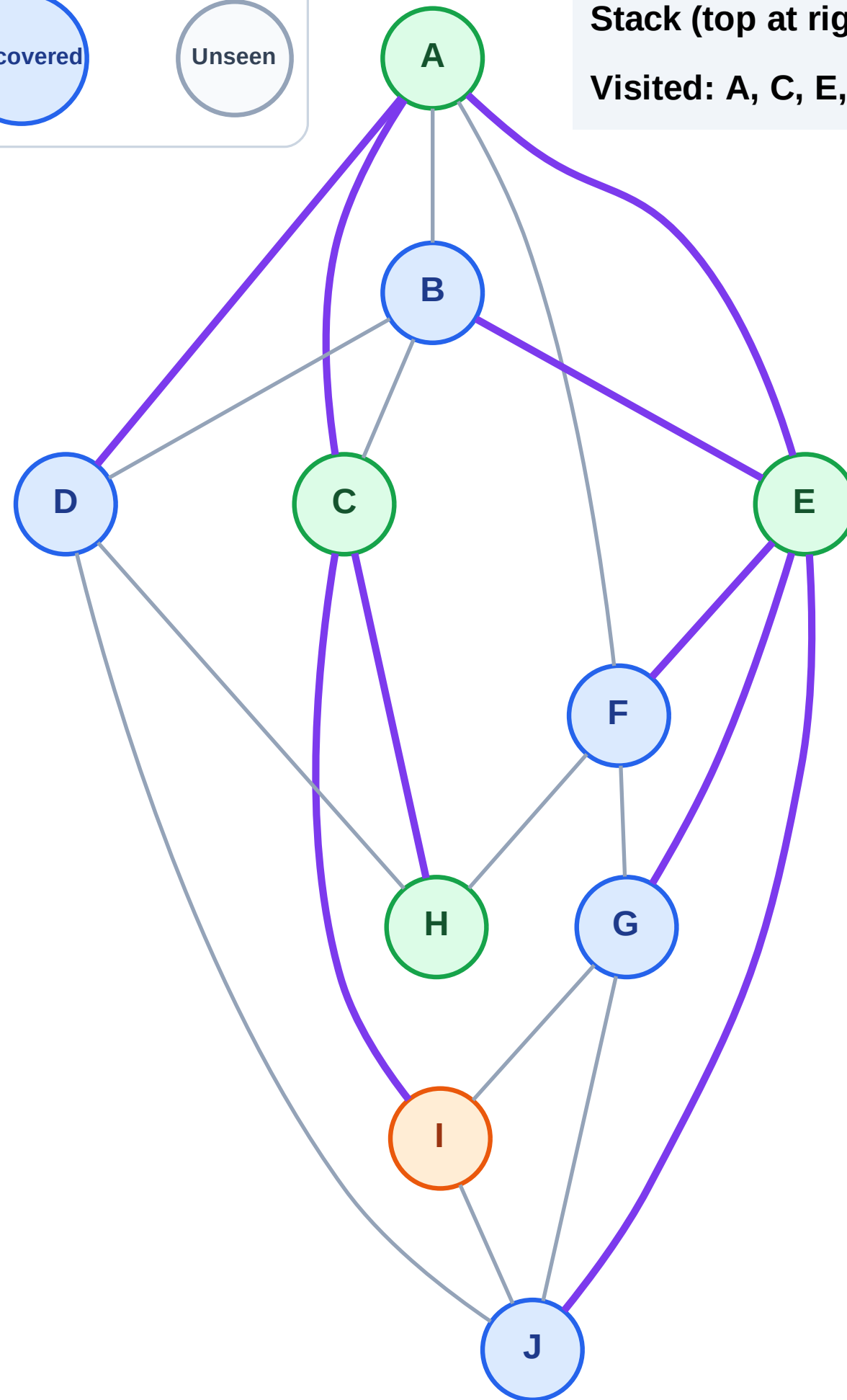
Discovered



Unseen

Stack (top at right): J | G | F | B | D

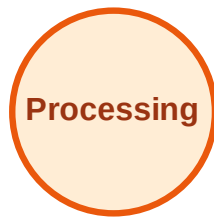
Visited: A, C, E, H



DFS Traversal

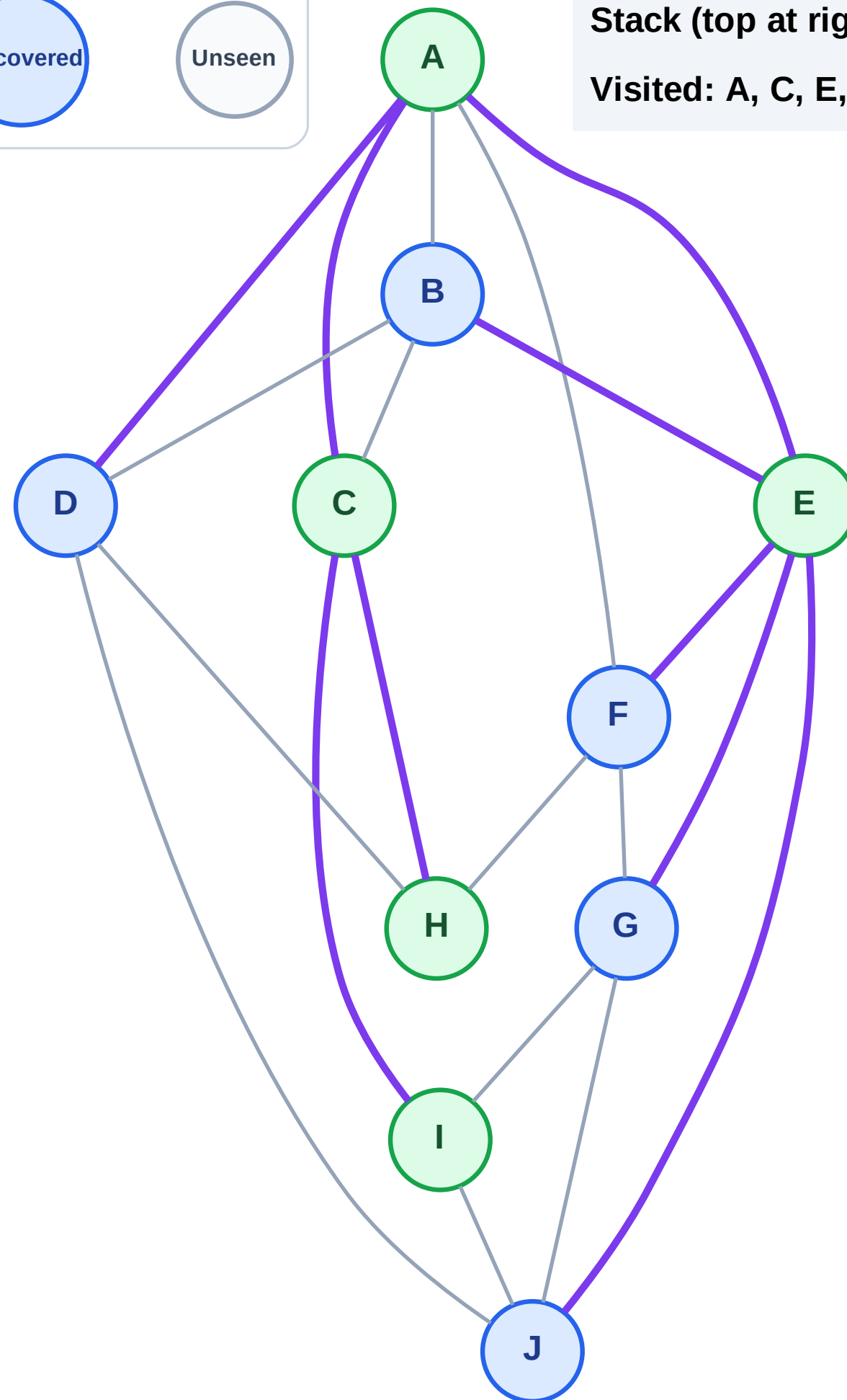
Step 11: No new neighbors from I

Legend



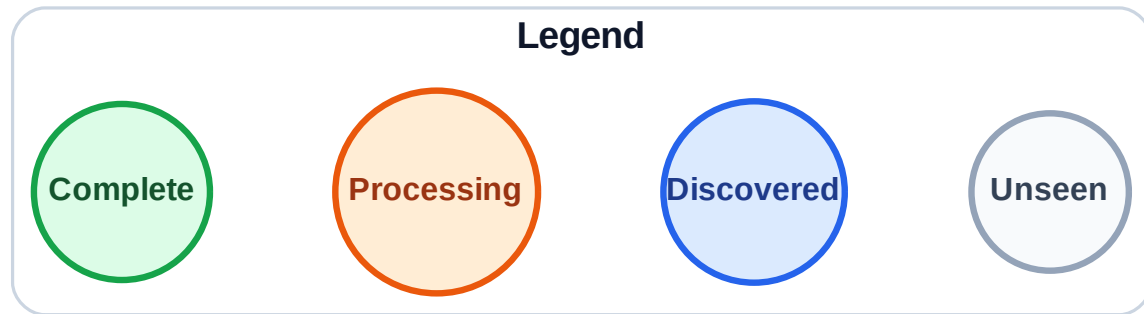
Stack (top at right): J | G | F | B | D

Visited: A, C, E, H, I

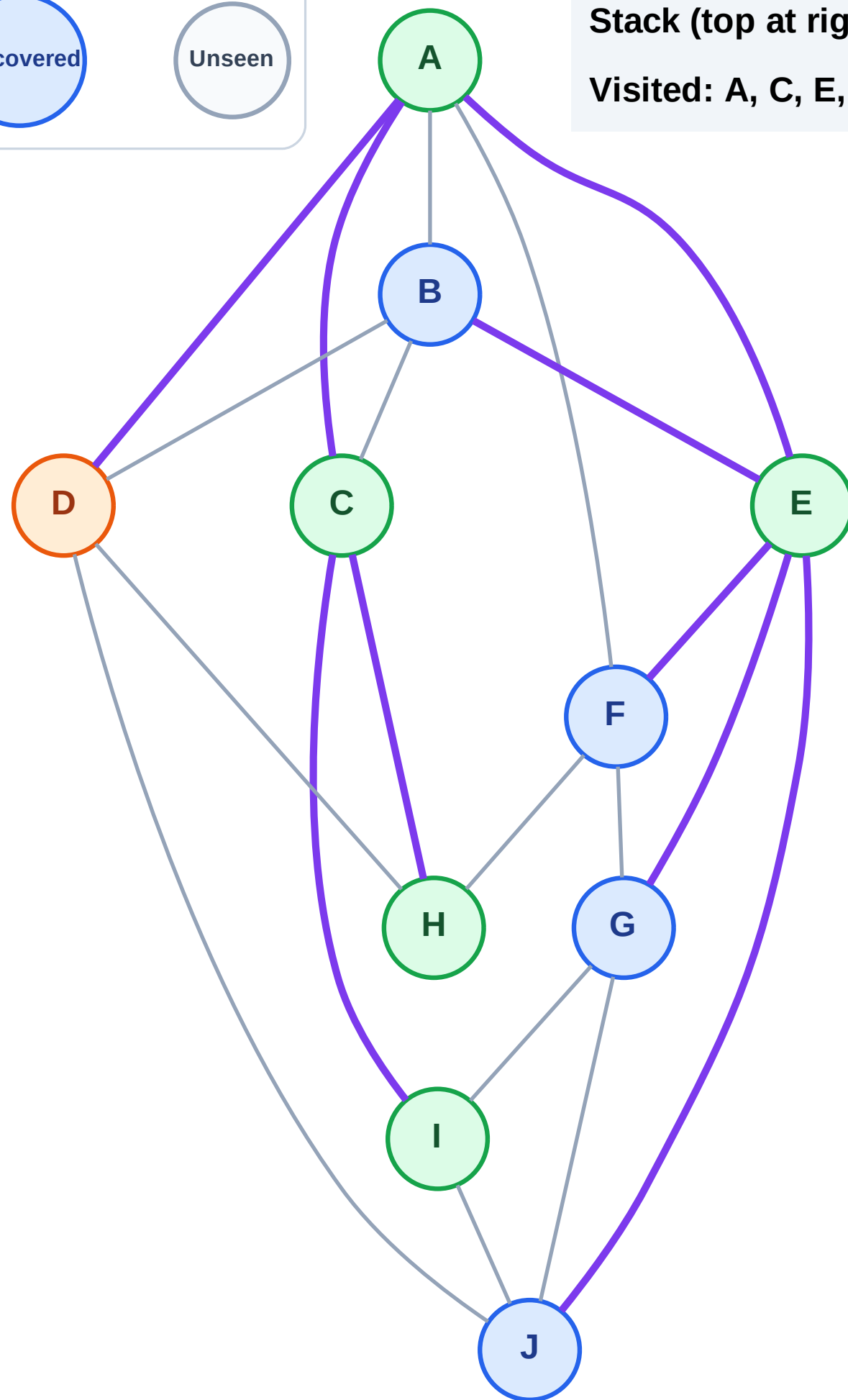


DFS Traversal

Step 12: Process node D



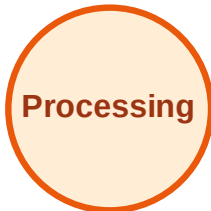
Stack (top at right): J | G | F | B
Visited: A, C, E, H, I



DFS Traversal

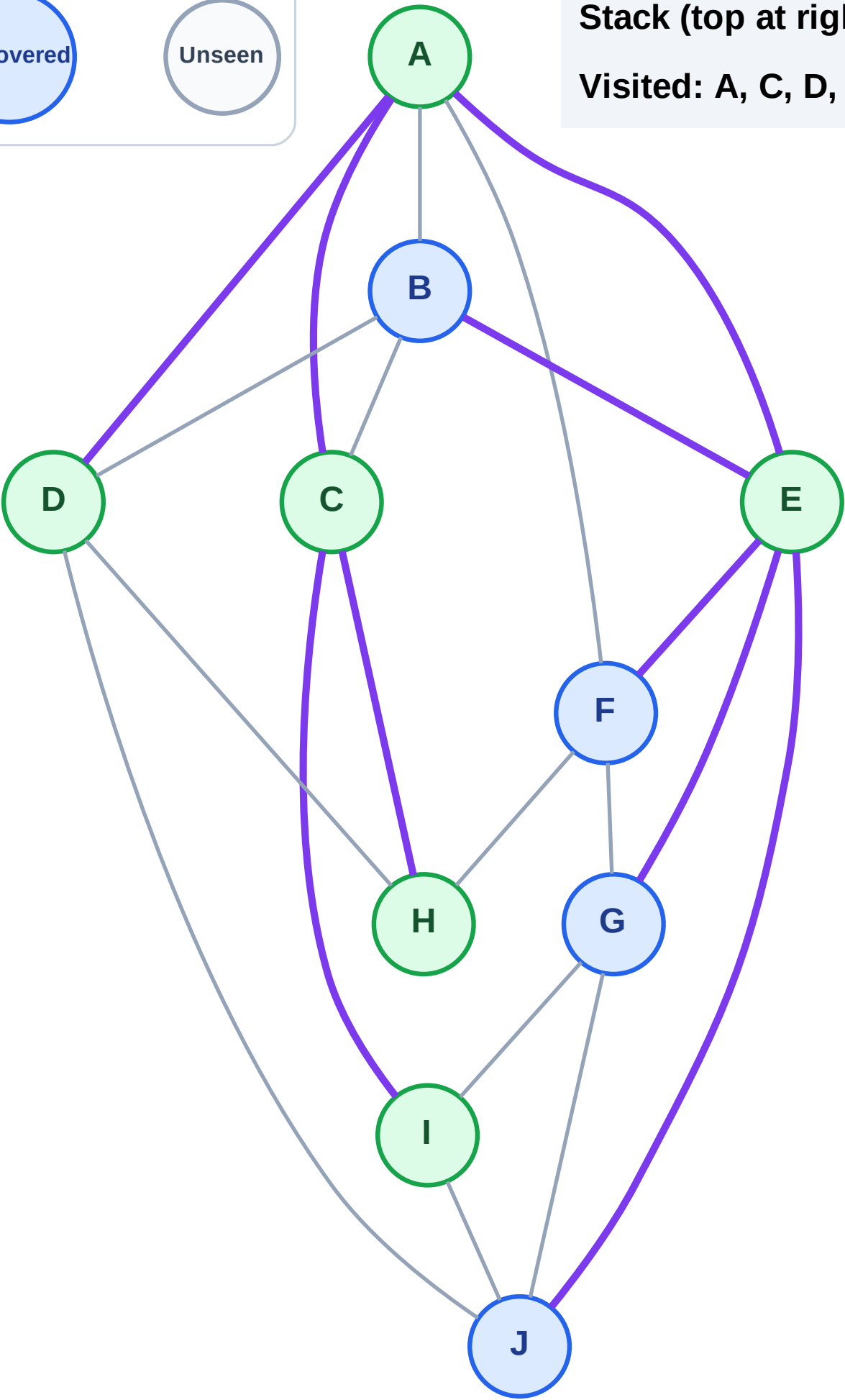
Step 13: No new neighbors from D

Legend



Stack (top at right): J | G | F | B

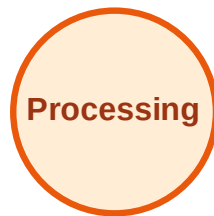
Visited: A, C, D, E, H, I



DFS Traversal

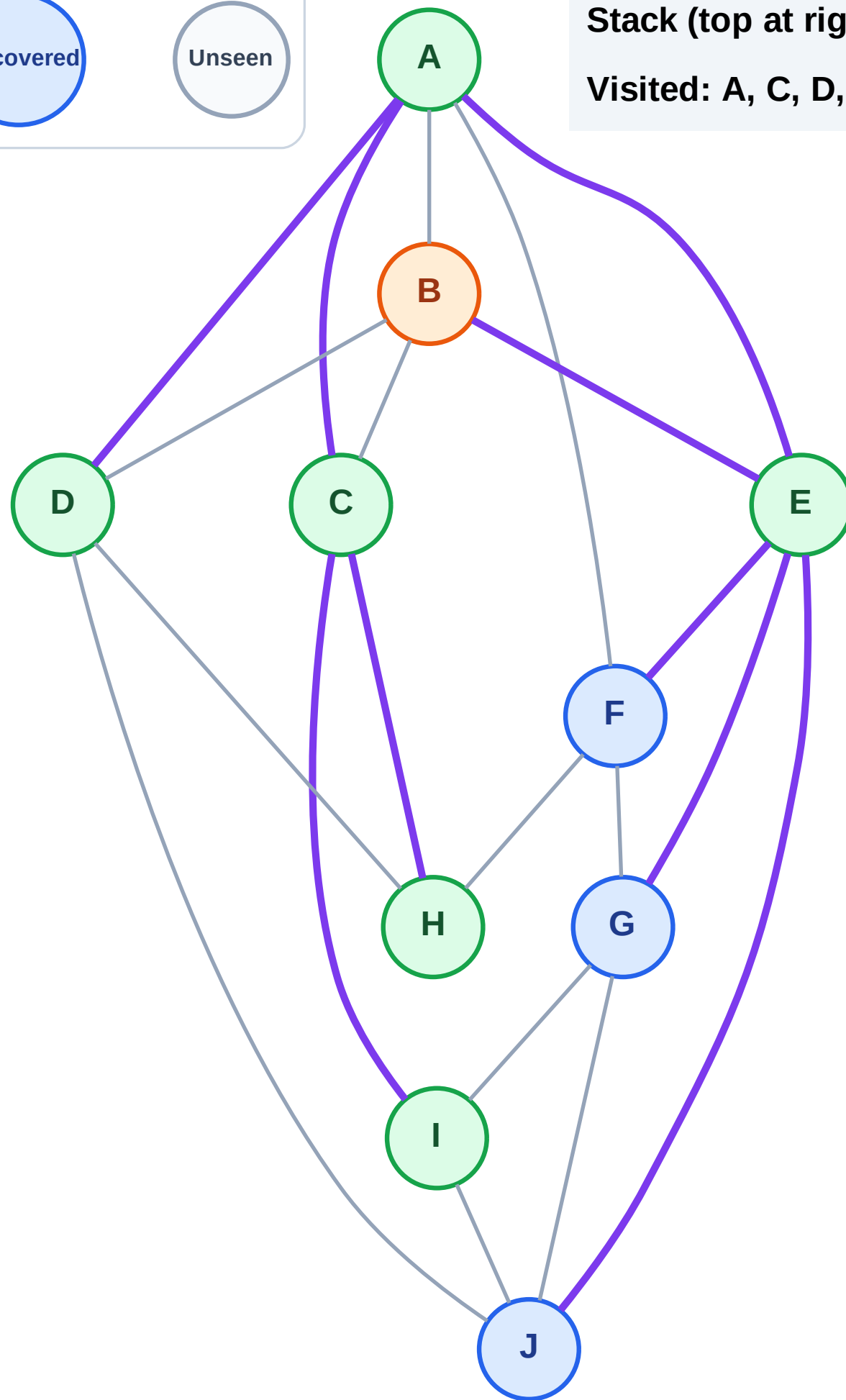
Step 14: Process node B

Legend



Stack (top at right): J | G | F

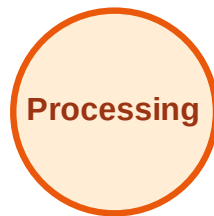
Visited: A, C, D, E, H, I



DFS Traversal

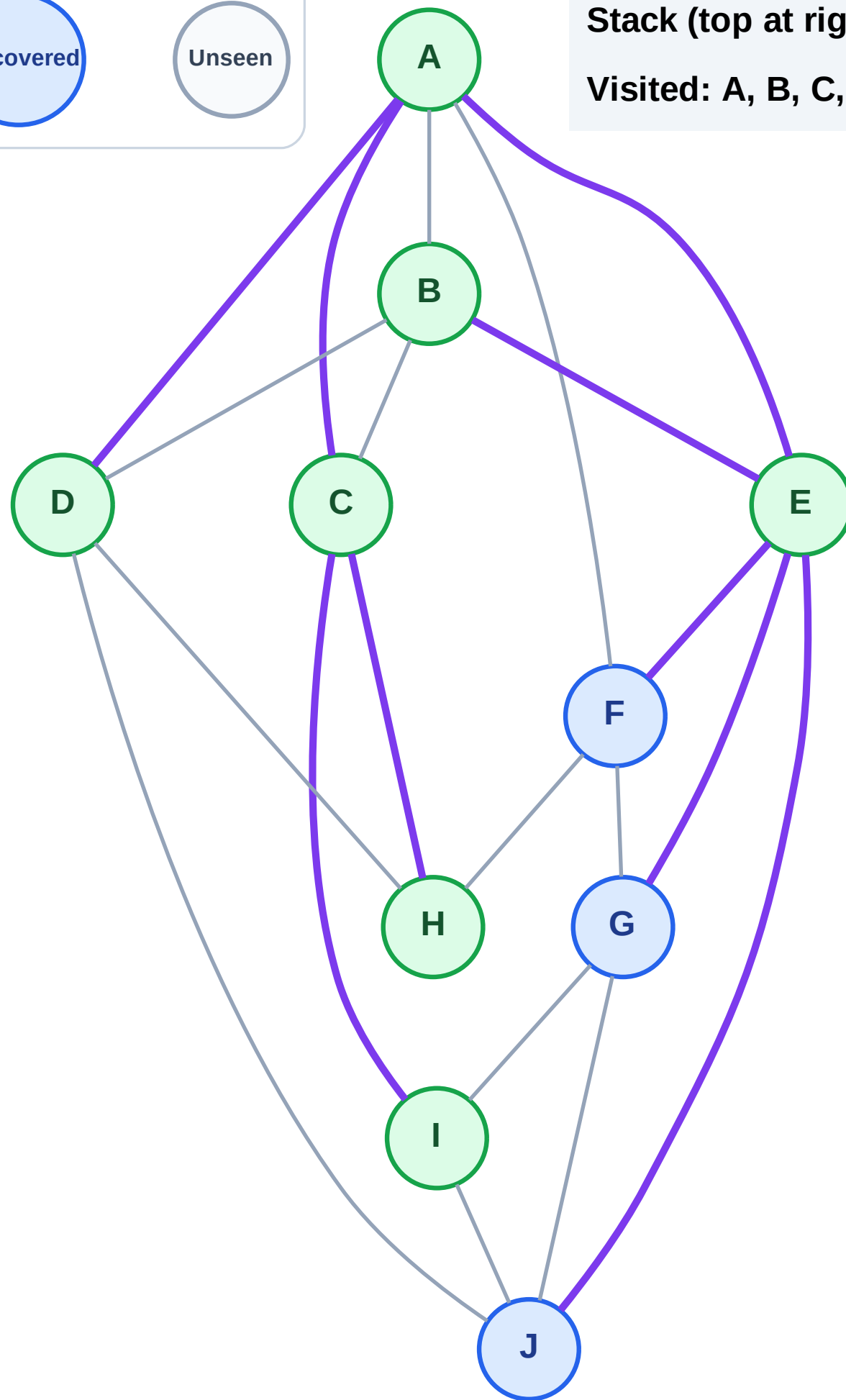
Step 15: No new neighbors from B

Legend



Stack (top at right): J | G | F

Visited: A, B, C, D, E, H, I



DFS Traversal

Step 16: Process node F

Legend

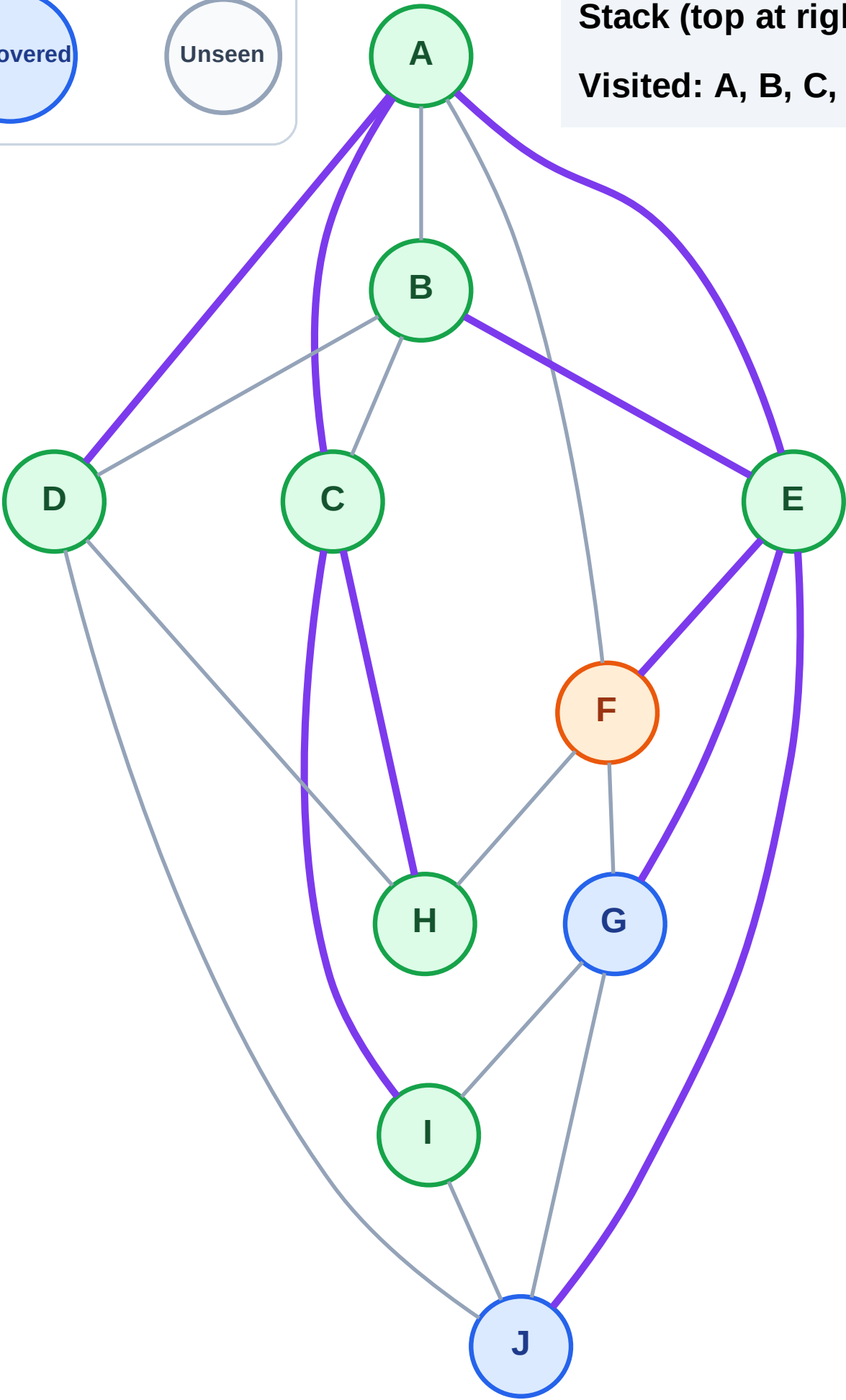
Complete

Processing

Discovered

Unseen

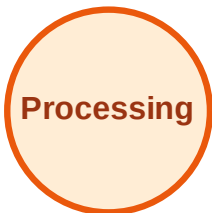
Stack (top at right): J | G
Visited: A, B, C, D, E, H, I



DFS Traversal

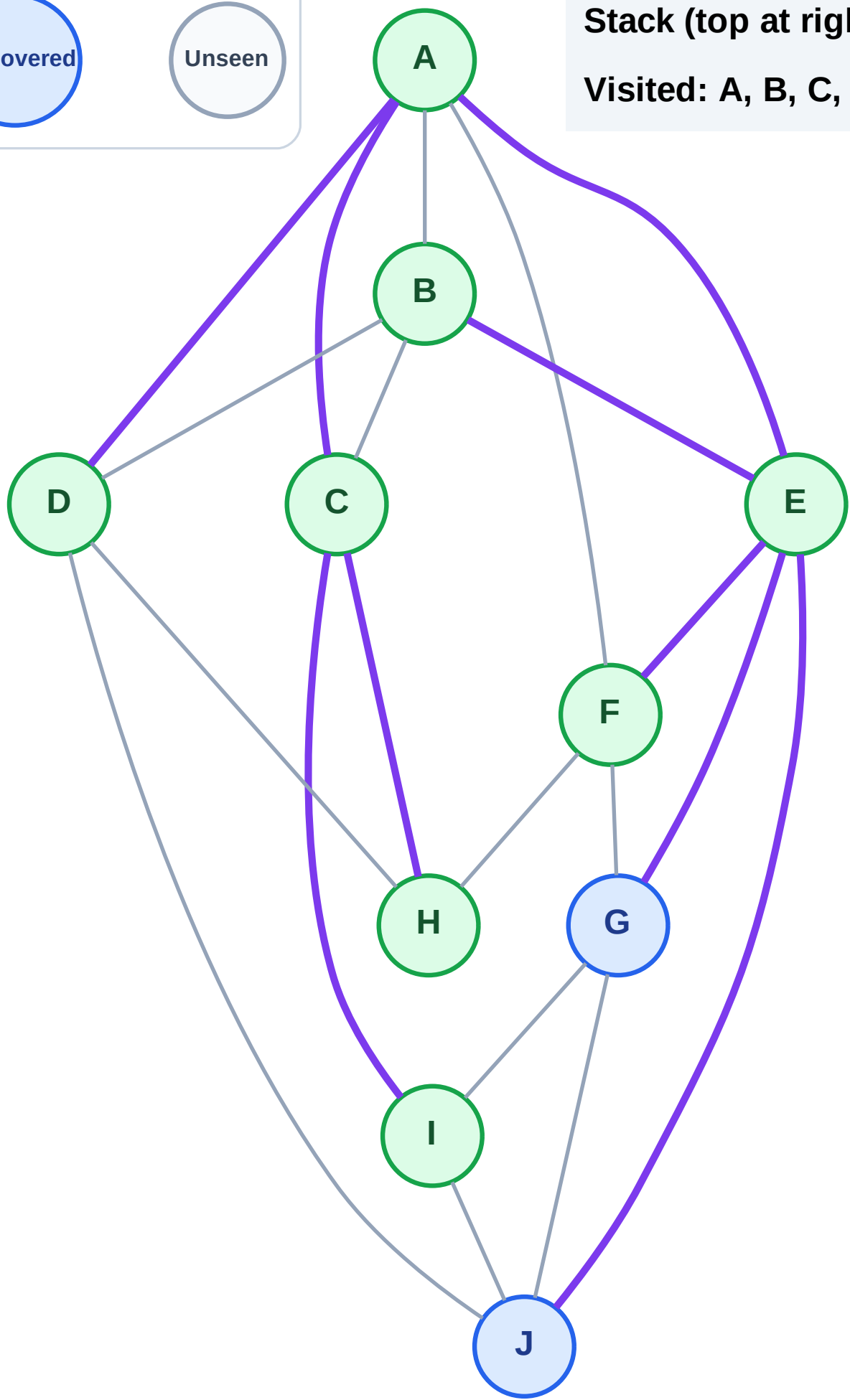
Step 17: No new neighbors from F

Legend



Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



DFS Traversal

Step 18: Process node G

Legend

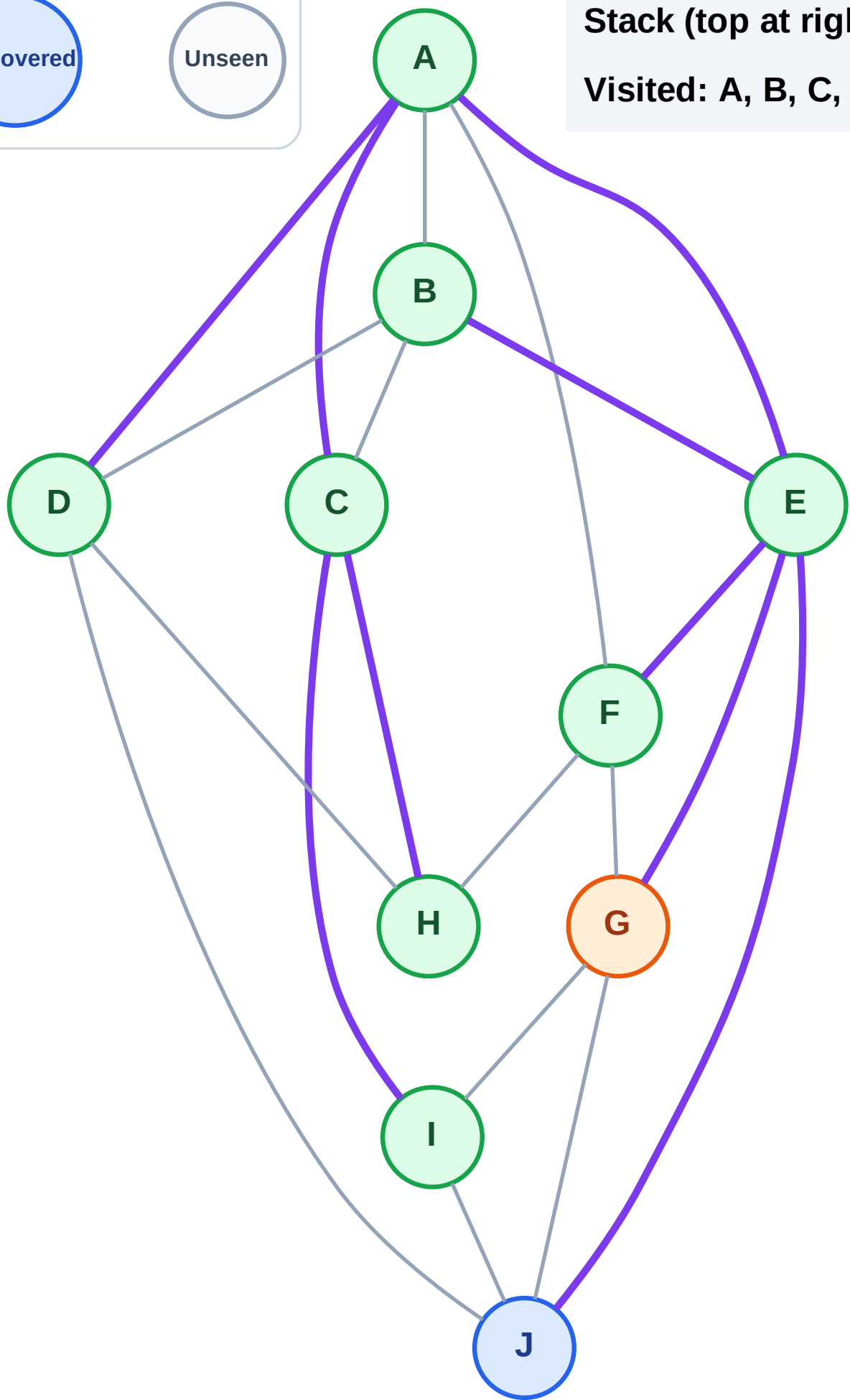
Complete

Processing

Discovered

Unseen

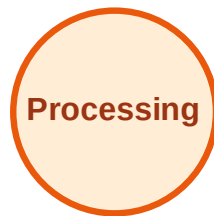
Stack (top at right): J
Visited: A, B, C, D, E, F, H, I



DFS Traversal

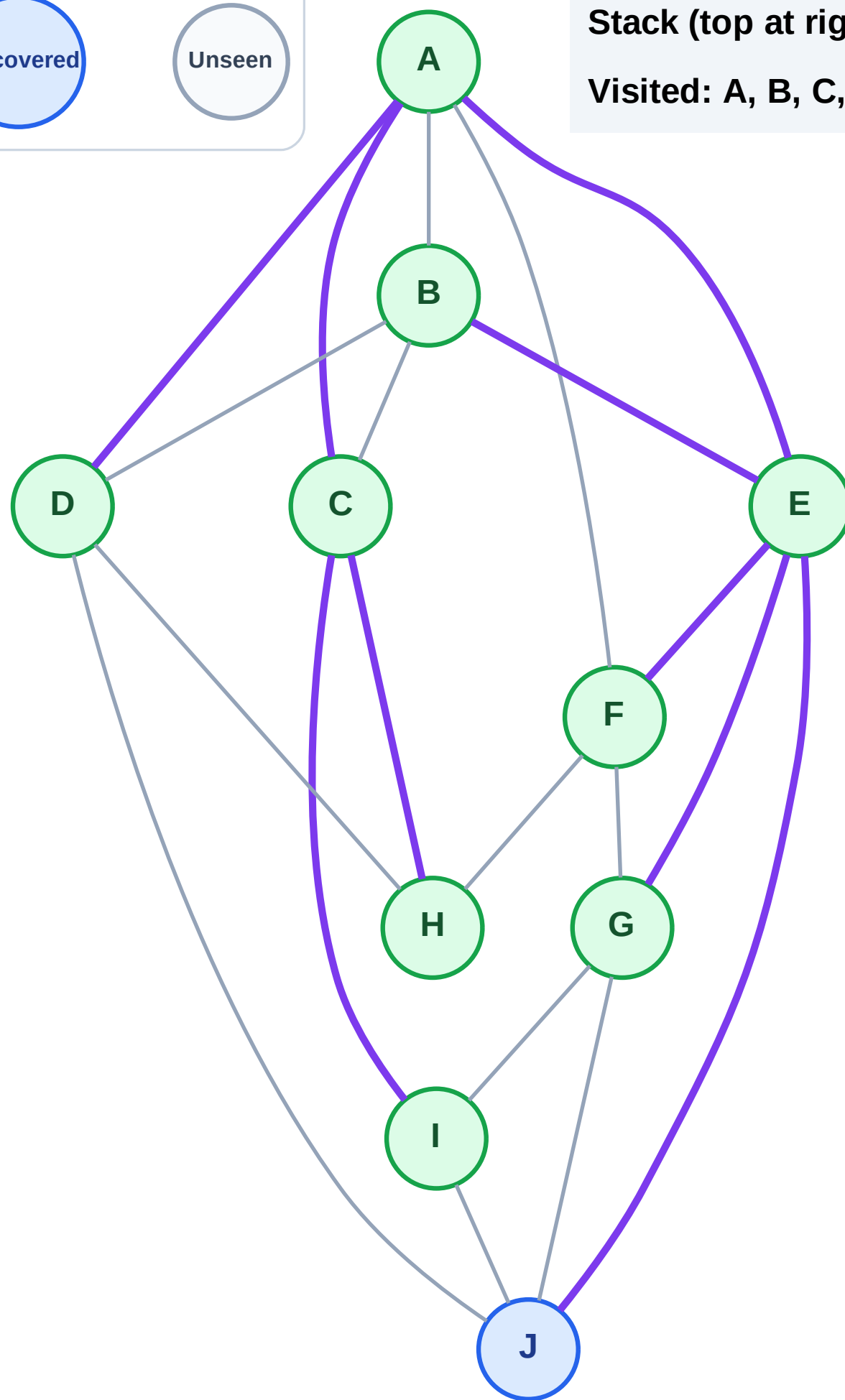
Step 19: No new neighbors from G

Legend



Stack (top at right): J

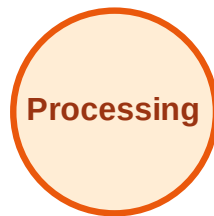
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

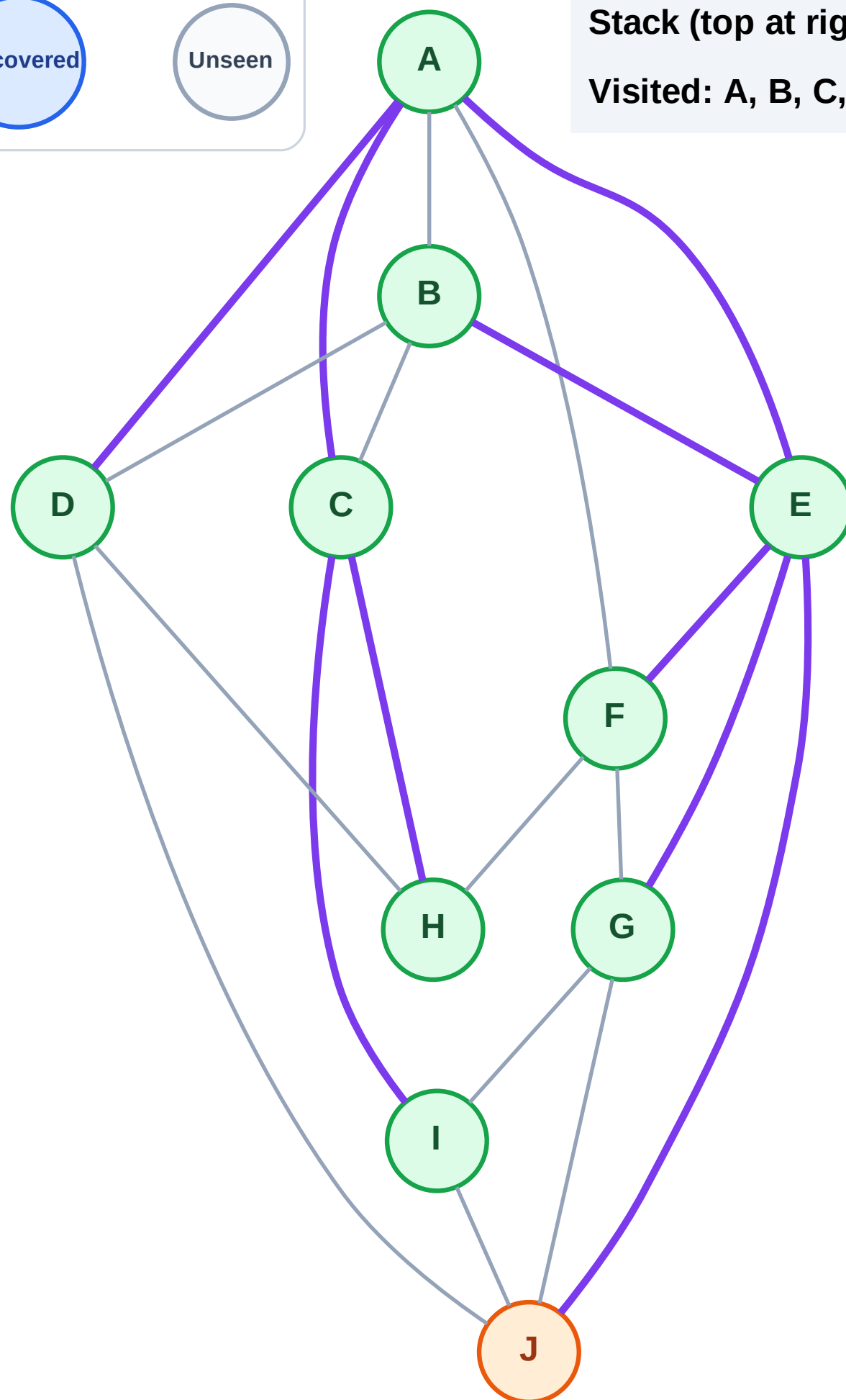
Step 20: Process node J

Legend



Stack (top at right): (empty)

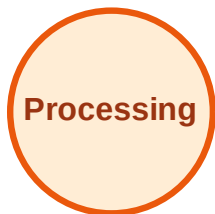
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

